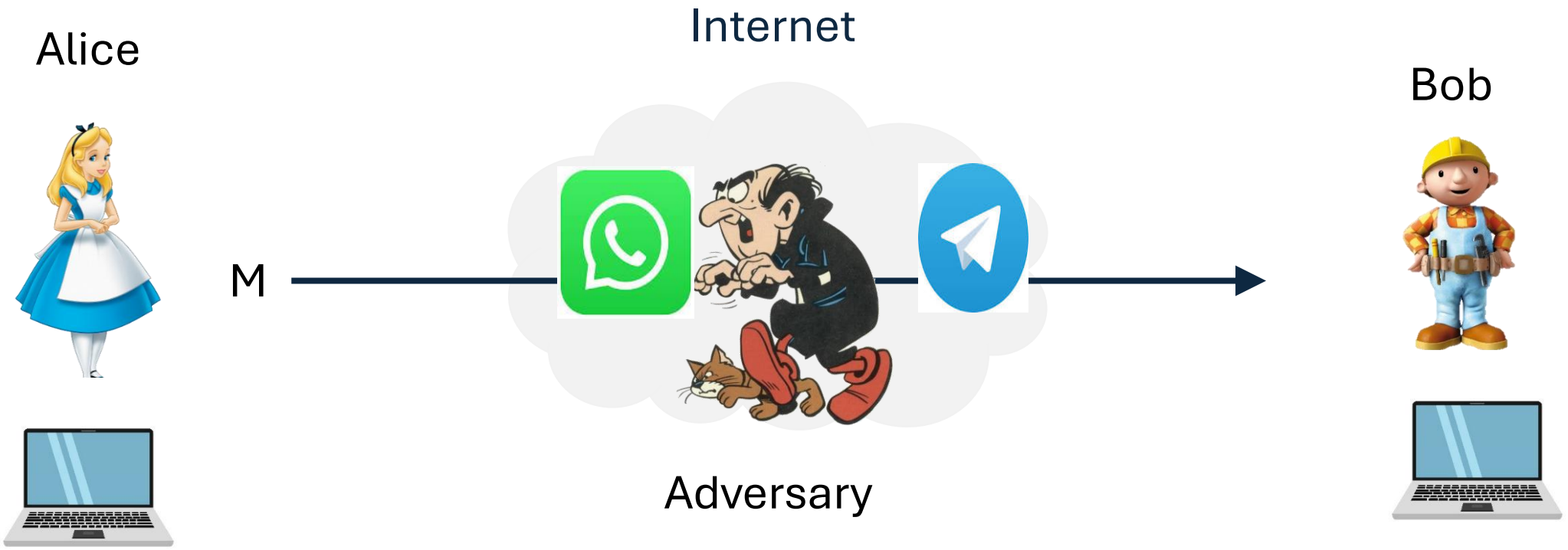
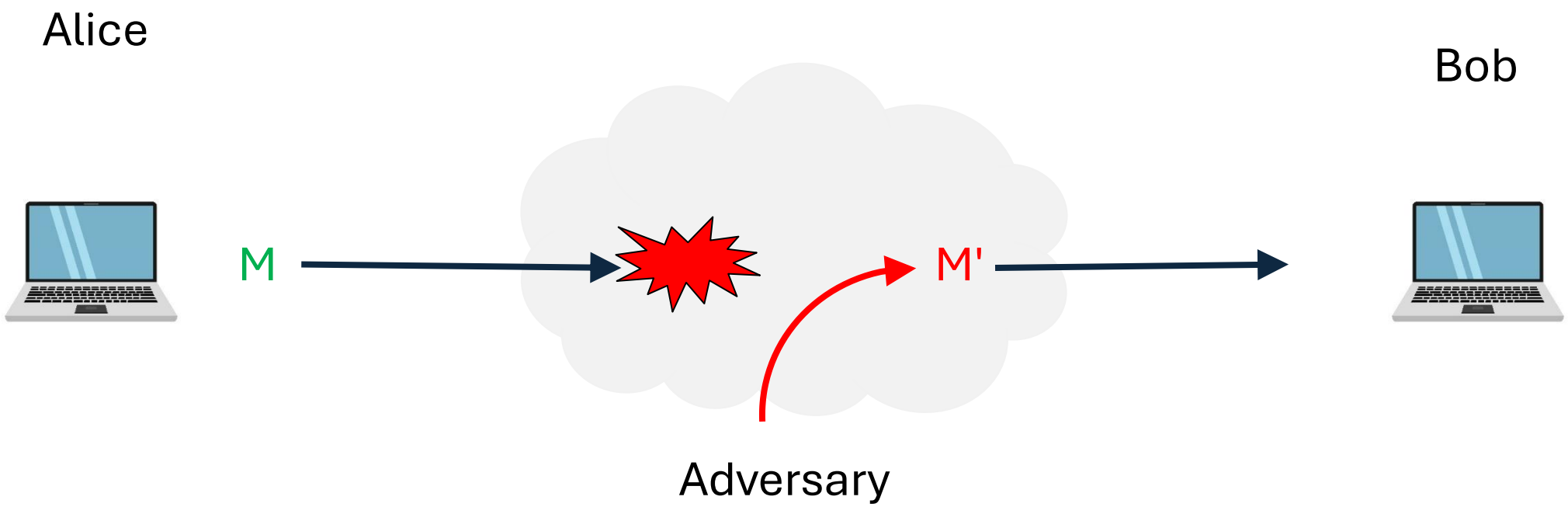
Week 2– Şifreleme Algoritmaları

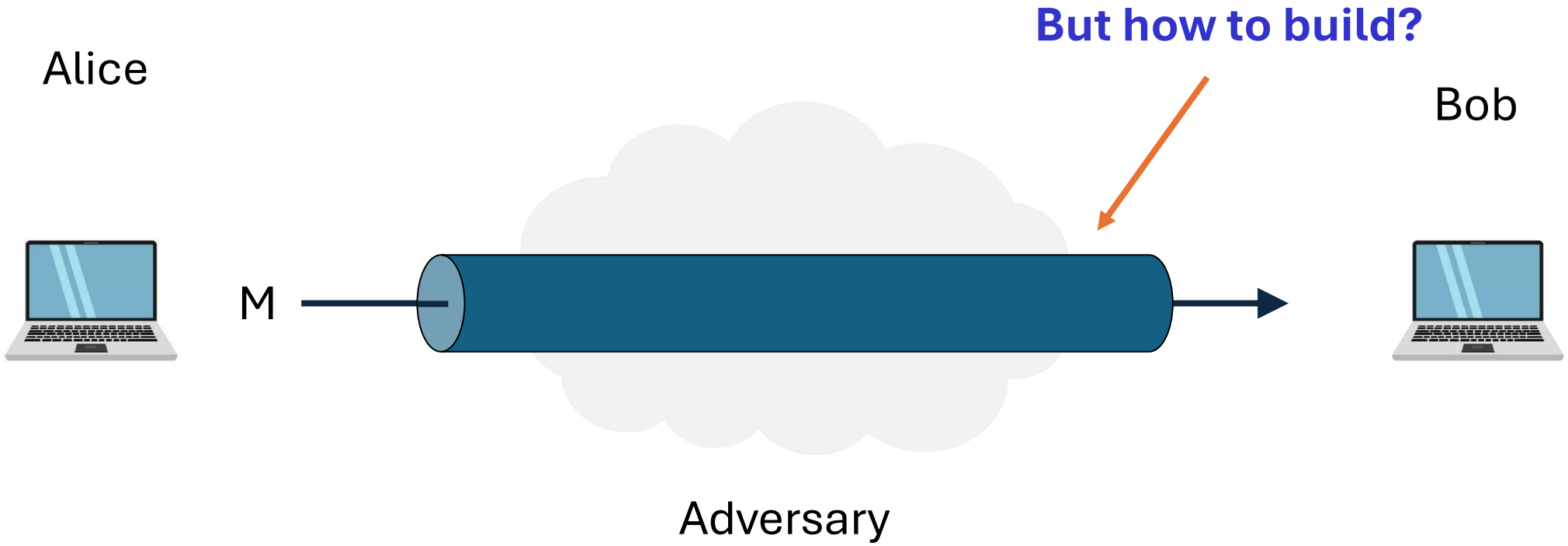

Alice
Internet
Bob
M
Adversary



Security goals(Güvenlik hedefleri):

- Data privacy(Veri Gizliliği): Mesaj okunamasın.
- Data integrity (V. Bütünlüğü): Mesaj değiştirilemesin.
- Data authenticity (V. Kimlik Doğrulaması): message M really originated from Alice

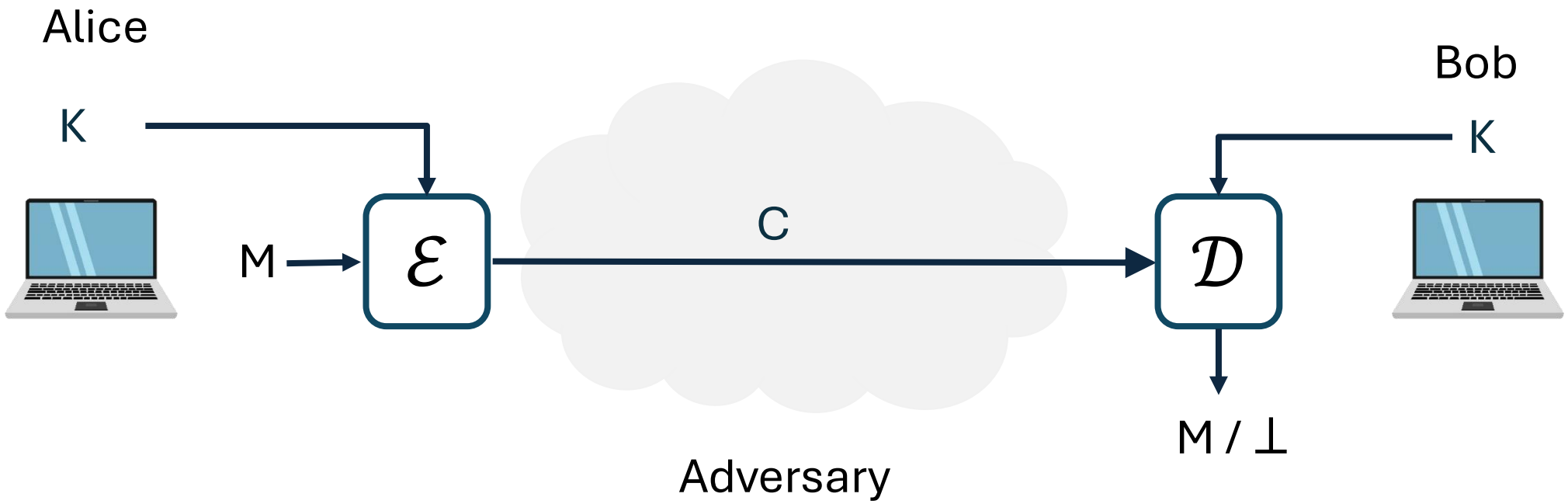
Ideal solution: secure channels



Güvenlik hedefleri :

- **Gizlilik**: Okuma yasağı ✓
- **Bütünlük:** Değiştirme yasağı ✓
- **Kimlik :** Kaynağın doğruluğu ✓

Creating secure channels: encryption schemes

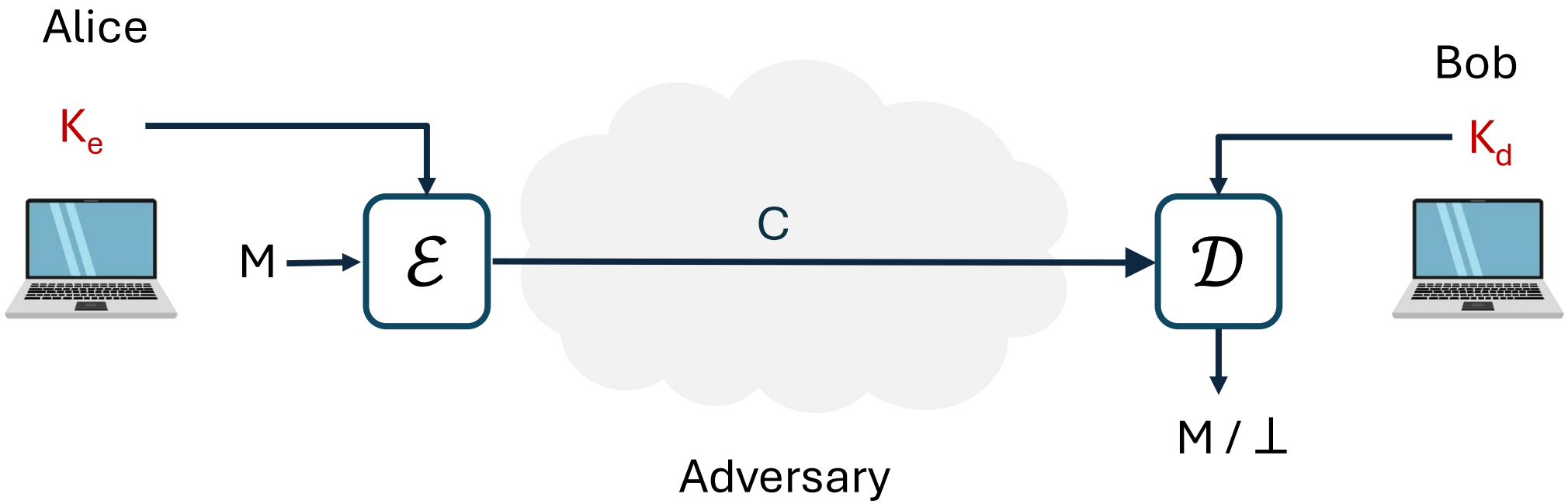


$\mathcal{E}$ **:** Şifreleme algoritması (genel/açık)

K :Şifreleme / Şifre çözme anahtarı (gizli)

$\mathcal{D}$ **:** Şifre çözme algoritması (genel/açık))

Creating secure channels: encryption schemes



$\mathcal{E}$: Şifreleme algoritması (genel/açık)

$\mathcal{D}$: Şifre çözme algoritması (genel/açık)

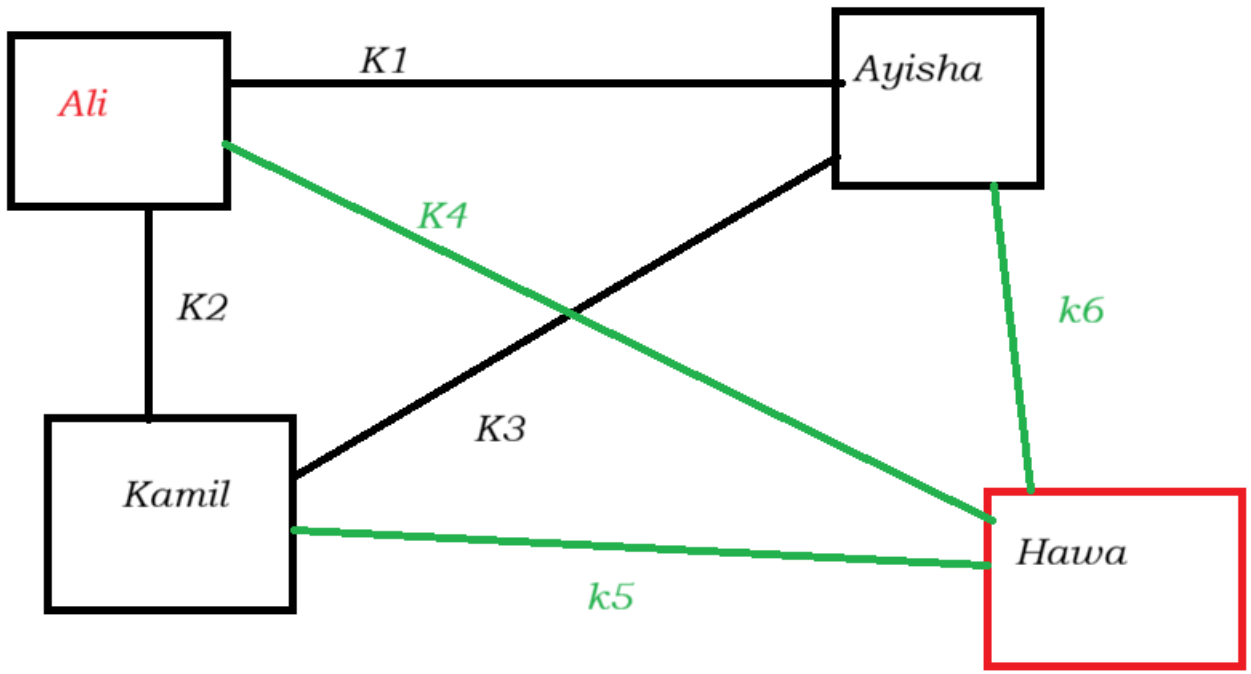
K_e : Şifreleme anahtarı (genel/açık)

K_d : Şifre çözme anahtarı (gizli)

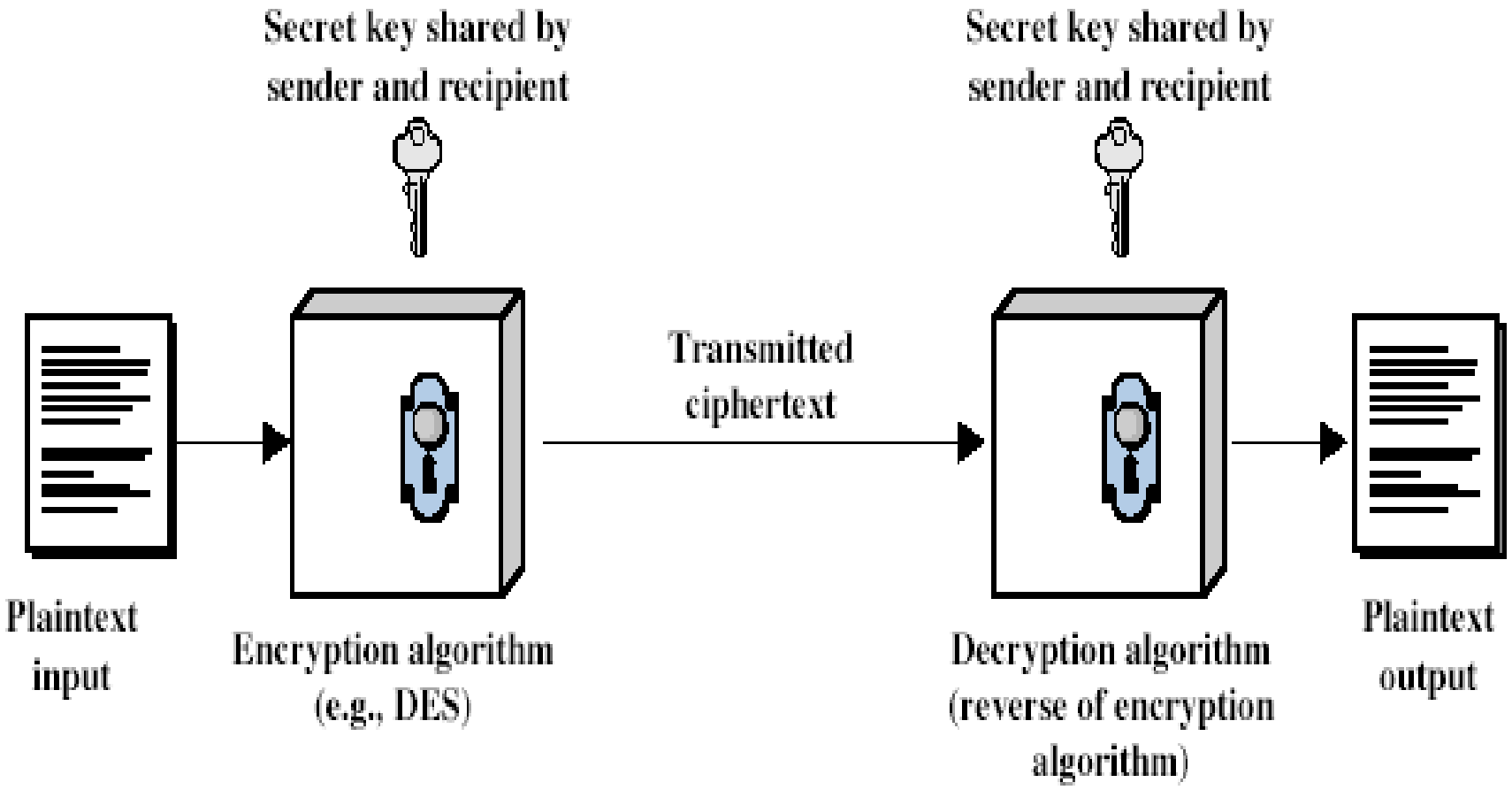
Şifreleme Algoritmalarının Sınıflandırılması

- **Simetrik Şifreler (Symmetric Ciphers):**
  - Şifreleme ve şifre çözme için **aynı gizli anahtar** kullanılır
  - Gönderici ve alıcı anahtarı **güvenli bir şekilde paylaşmak** zorundadır
- **Asimetrik Şifreler (Asymmetric Ciphers):**
  - Şifreleme ve şifre çözme için **farklı anahtarlar** kullanılır
  - Açık anahtar (Public Key) ve gizli anahtar (Private Key) **matematiksel olarak ilişkilidir**
  - **Açık anahtar:** şifreleme için kullanılır
  - **Gizli anahtar:** şifre çözme için kullanılır

Symmetric key....



Simetrik Şifreleme Modeli (Symmetric Cipher Model)



Gereksinimler (Requirements) – Simetrik Şifreleme İçin

- Simetrik Şifrelemenin Güvenli Kullanımı İçin İki Gereklilik
  - Güçlü bir şifreleme algoritması
  - Sadece gönderici ve alıcının bildiği gizli anahtar
    - C= $E_K(M)$ *(Şifreleme)*
    - M= $D_K(C)$ *(Şifre çözme)*
- Şifreleme algoritmasının bilindiği varsayılır
- Bu durum, anahtarın güvenli bir şekilde dağıtılmasını gerektirir

Klasik Substitüsyon (Yerine Koyma) Şifreleri

- Düz metindeki harfler, başka harflerle, sayılarla veya sembollerle değiştirilir
- Düz metin **bit dizisi** olarak görülür ve **yerine koyma**, düz metin bit desenlerini şifreli metin bit desenleri ile değiştirir

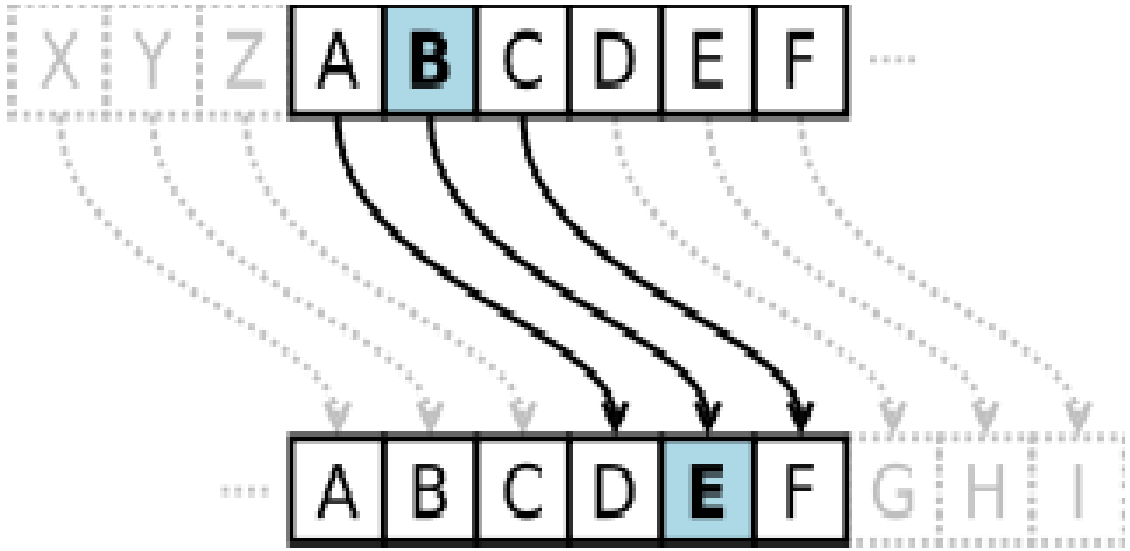
```
meet me after the toga party
PHHW PH DIWHU WKH WRJD SDUWB
```

- https://raw.org/tool/caesar-cipher/

Shift cipher

Cesar:

k = 3, 4,5.....



Symmetric encryption – syntax

$\Pi = (\mathcal{E}, \mathcal{D})$

$\mathcal{E} : \mathcal{K} \times \mathcal{M} \to \mathcal{C}$

$\mathcal{E}(K, M) = \mathcal{E}_K(M) = C$

$\mathcal{D}: \mathcal{K} \times \mathcal{C} \to \mathcal{M}$

$\mathcal{D}(K, C) = \mathcal{D}_K(C) = M$

Examples:

$\mathcal{K} = \{0,1\}^{128}$	$\mathcal{M} = \{0,1\}^*$	$\mathcal{C} = \{0,1\}^*$
$\mathcal{K} = \{0,1\}^{128}$	$\mathcal{M} = \{A, B, \dots, Z\}$	$\mathcal{C} = \{A, B, \dots, Z\}$
$\mathcal{K} = \{0,1\}^{128}$	$\mathcal{M} = \{\text{YES}, \text{NO}\}$	$\mathcal{C} = \{0,1\}^*$
$\mathcal{K} = \{1, \dots, p\}$	$\mathcal{M} = \{A, B, \dots, Z\}$	$\mathcal{C} = \{0,1\}^*$

- **Correctness requirement:**
- $\forall K \in \mathcal{K}, \forall M \in \mathcal{M}$:
  - $\mathcal{D}\big(K, \mathcal{E}(K, M)\big) = M$

Possible privacy security goals:

- Hard to recover M from C
- Hard to recover K from C
- Hard to learn one bit of M from C
- Hard to learn parity of M from C
- …

Caesar Cipher

- Define transformation as:

```
a b c d e f g h i j k l m n o p q r s t u v
 w x y z
D E F G H I J K L M N O P Q R S T U V W X Y
 Z A B C
```

- Mathematically give each letter a number

```
a b c d e f g h i j k  l  m
0 1 2 3 4 5 6 7 8 9 10 11 12
n  o  p  q  r  s  t  u  v  w  x  y  Z
13 14 15 16 17 18 19 20 21 22 23 24 25
```

- Then have Caesar cipher as:

  $C = E(p) = (p + k) \bmod (26)$

  $p = D(C) = (C - k) \bmod (26)$

Sezar Şifresinin Kriptanalizi (Caesar Cipher)

- Sezar şifresinde sadece 25 olası kaydırma (shift) vardır
    - A → B, ..., Z
- Şifreli metin verildiğinde, tüm kaydırmaları deneyerek düz metni bulabilirsiniz
- Düz metni tanıyacak bir yöntem gerekir (ör. mantıklı İngilizce kelimeler)
- **Örnek:**
    - Şifreli metin: "GCUA VQ DTGCM"
    - Tüm 25 kaydırmayı deneyerek doğru kaydırmayı bulursunuz → "FLAG TO BREAK"

Ceasar cipher(Sezar Şifrelemesi)

- a b c d e f g h i j k l m n o p q r s t u v w x y z

...

- a b c d e f g h i j k l m n o p q r s t u v w x y z a

```
in the far distance a helicopter skimmed down between
the roofs, hovered for an instant like a bluebottle,
and darted away again with a curving flight. It was
the police patrol, snooping into people's windows

lq wkh idu glvwdqfh d kholfrswhu vnlpphg grzq ehwzhhq
wkh urriv, kryhuhg iru dq lqvwdqw olnh d eoxherwwoh,
dqg gduwhg dzdb djdlq zlwk d fxuylqj ioljkw. Lw zdv
wkh srolfh sdwuro, vqrrslqj lqwr shrsoh'v zlqgrzv
```

Ceasar cipher (ROT-13)

- a b c d e f g h i j k l m n o p q r s t u v w x y z

- a b c d e f g h i j k l m n o p q r s t u v w x y z

```
in the far distance a helicopter skimmed down between
the roofs, hovered for an instant like a bluebottle,
and darted away again with a curving flight. It was
the police patrol, snooping into people's windows

va gur sne qvfgnapr n uryvpbcgre fxvzzrq qbja orgjrra
gur ebbsf, ubirerq sbe na vafgnag yvxr n oyhrobggyr,
naq qnegrq njnl ntnva jvgu n pheivat syvtug. Vg jnf
gur cbyvpr cngeby, fabbcvat vagb crbcyr'f jvaqbjf
```

Attacking ROT-K

$|\mathcal{K}| = 26$

K	M
0	va gur sne qvfgnapr n uryvpbcgre …

C = va gur sne qvfgnapr n uryvpbcgre …

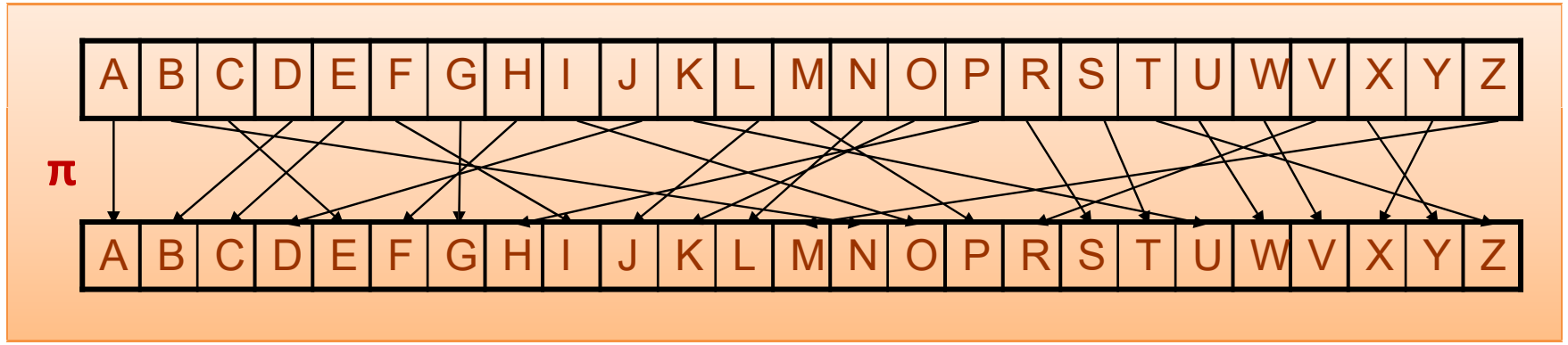
Conclusion: key space must be large enough!

Substitution cipher

- a b c d e f g h i j k l m n o p q r s t u v w x y z
- …
- a b c d e f g h i j k l m n o p q r s t u v w x y z
-

Substitution cipher

$\mathbb{M}$ = words over alphabet {A,...,Z} ≈ {0,...,25}
$\mathbb{K}$ = a set of permutations of {0,...,25}



$$Enc_{\pi}(m_0,\ldots,m_n) = (\pi(m_0),\ldots, \pi(m_n))$$

$$Dec_{\pi}(c_0,\ldots,c_n) = (\pi^{-1}(c_0),\ldots, \pi^{-1}(c_n))$$

Substitution cipher

- a b c d e f g h i j k l m n o p q r s t u v w x y z
- …
- a b c d e f g h i j k l m n o p q r s t u v w x y z
-
-

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

Substitution cipher

a b c d e f g h i j k l m n o p q r s t u v w x y z

↕ ↕ ↕ ↕ ...

↕

in the far distance a helicopter skimmed down between the roofs, hovered for an instant like a bluebottle, and darted away again with a curving flight. It was the police patrol, snooping into people's windows

s x d y w q f m j k o i l g z b e n t u c p a r v h

jg umw qsn yjtusgdw s mwijdzbuwn tojllwy yzag xwuawwg umw nzzqt, mzpwnwy qzn sg jgtusgu ijow s xicwxzuuiw, sgy ysnuwy sasv sfsjg ajum s dcnpjgf qijfmu. ju ast umw bzijdw bsunzi, tgzzbjgf jguz bwzbiw't ajgyzat

How to break the substitution cipher?

Use **statistical patterns** of the language.

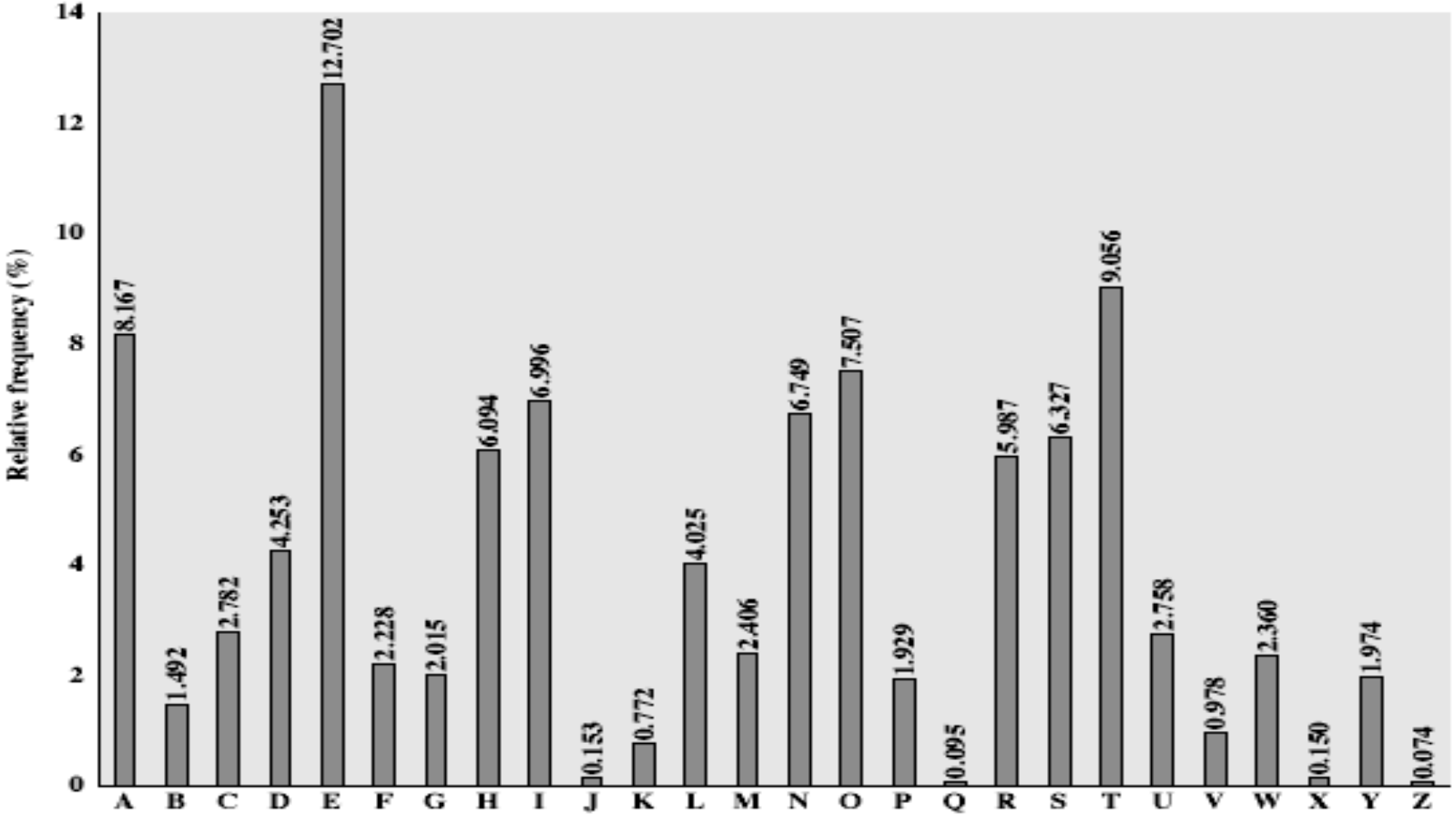
For example: the **frequency tables**.

Texts of **50** characters can usually be broken this way.

Letter	Frequency
E	0.127
T	0.097
I	0.075
A	0.073
N	0.067
S	0.067
R	0.064
H	0.049
C	0.045
L	0.040
D	0.031
P	0.030
Y	0.027
U	0.024
M	0.024
F	0.021
B	0.017
G	0.016
W	0.013
V	0.008
K	0.008
X	0.005
Q	0.002
Z	0.001
J	0.001

Figure 7 - Frequency Table

English Letter Frequencies



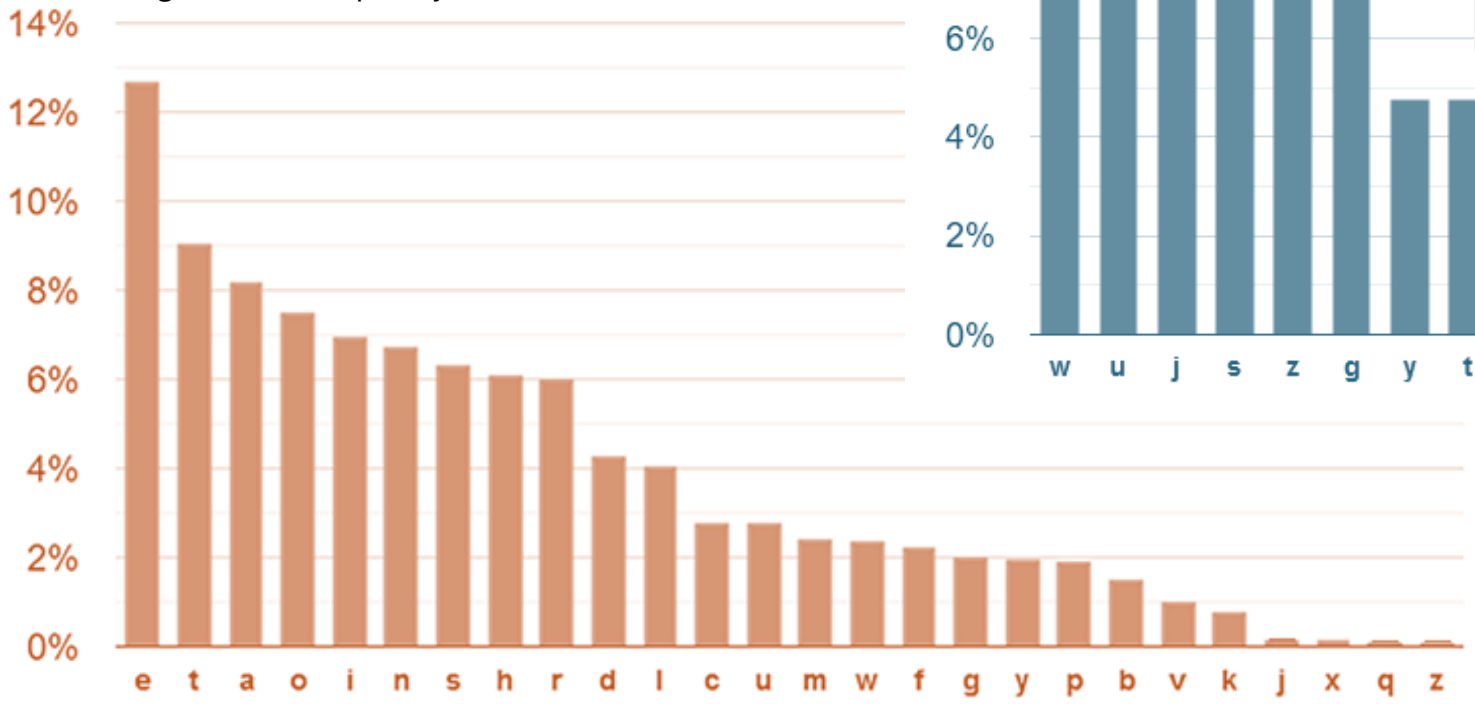
Note that all human languages have varying letter frequencies, though the number of letters and their frequencies varies.

Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
jg umw qsn yjtusgdw s mwijdzbuwn
tojllwy yzag xwuawwg umw nzzqt,
mzpwnwy qzn sg jgtusgu ijow s
xicwxzuuiw, sgy ysnuwy sasv sfsjg
ajum s dcnpjgf qijfmu. ju ast umw
bzijdw bsunzi, tgzzbjgf jguz
bwzbiw't ajgyzat
```

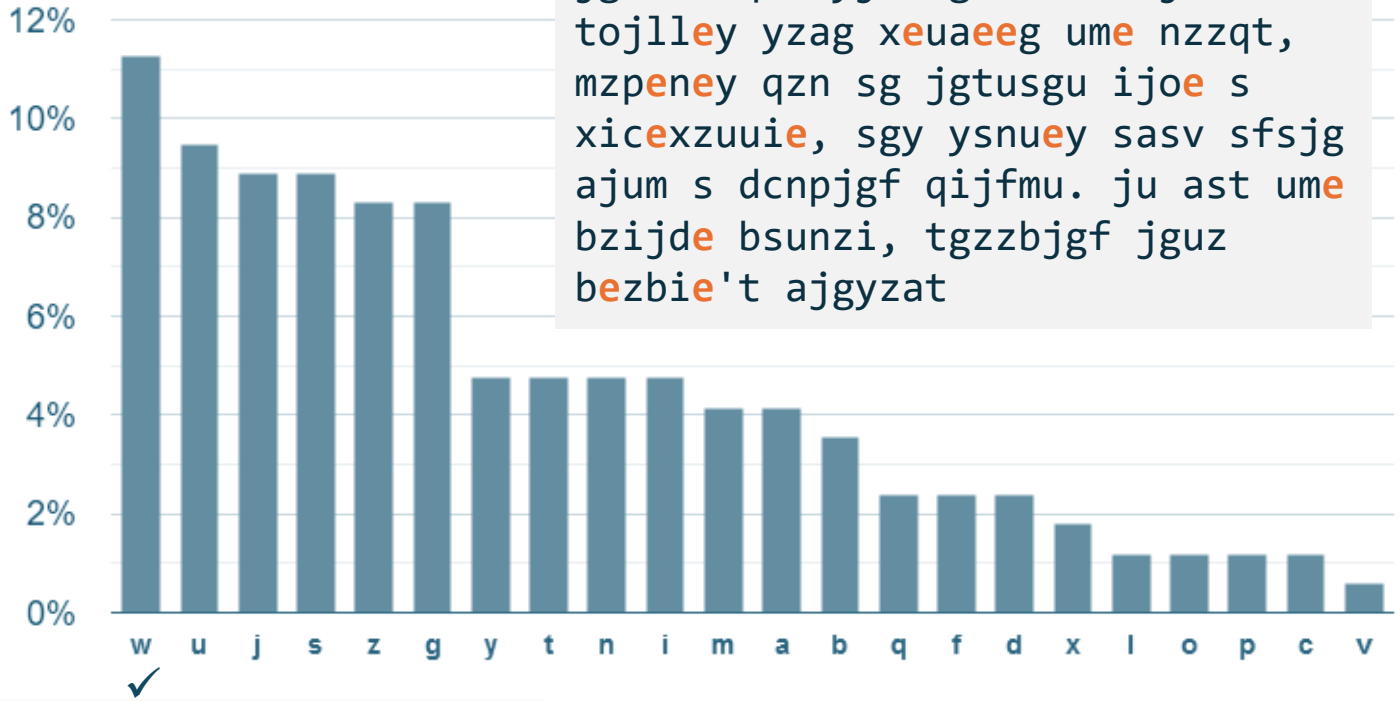
English letter frequency

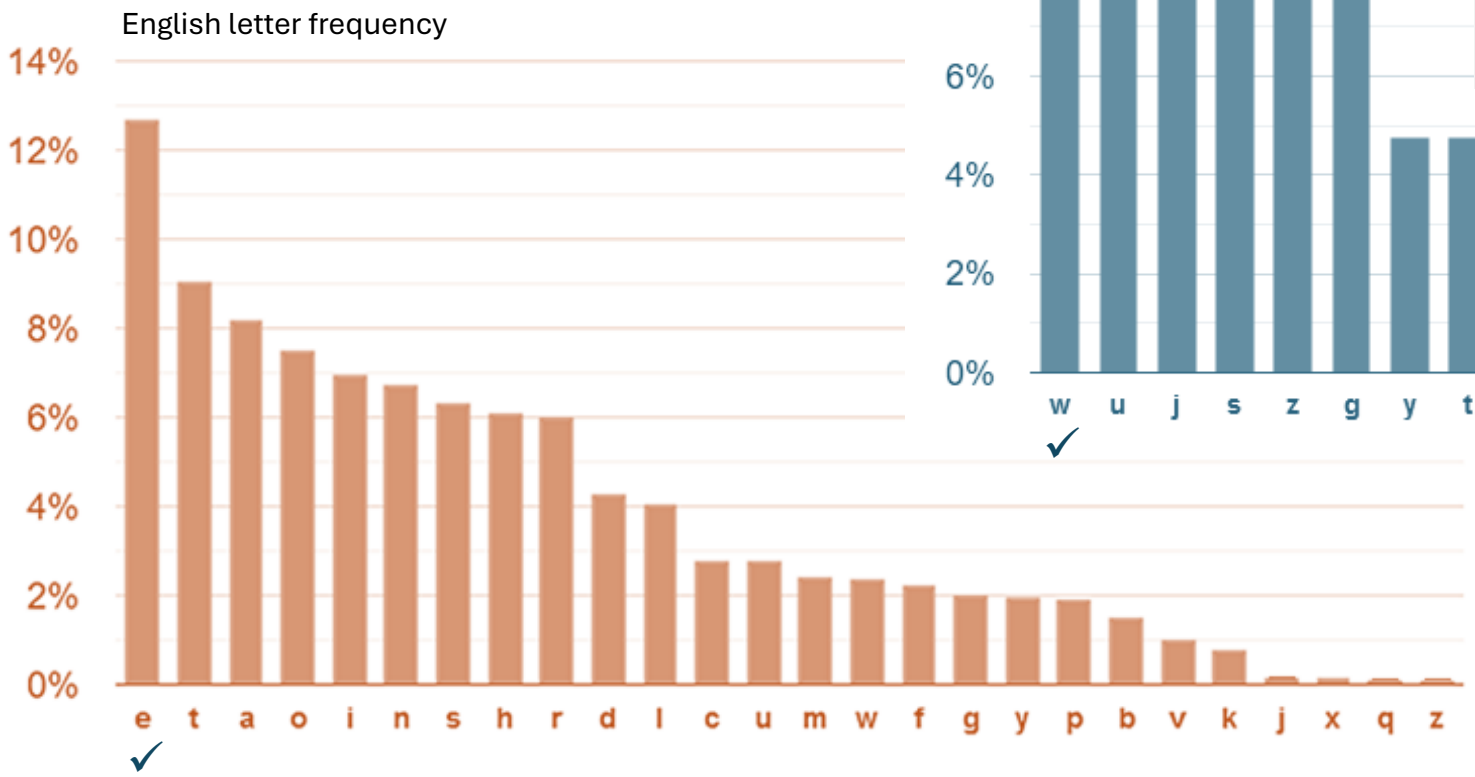


Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
jg ume qsn yjtusgde s meijdzbuen
tojlley yzag xeuaeeg ume nzzqt,
mzpeney qzn sg jgtusgu ijoe s
xicexzuuie, sgy ysnuey sasv sfsjg
ajum s dcnpjgf qijfmu. ju ast ume
bzijde bsunzi, tgzzbjgf jguz
bezbie't ajgyzat
```

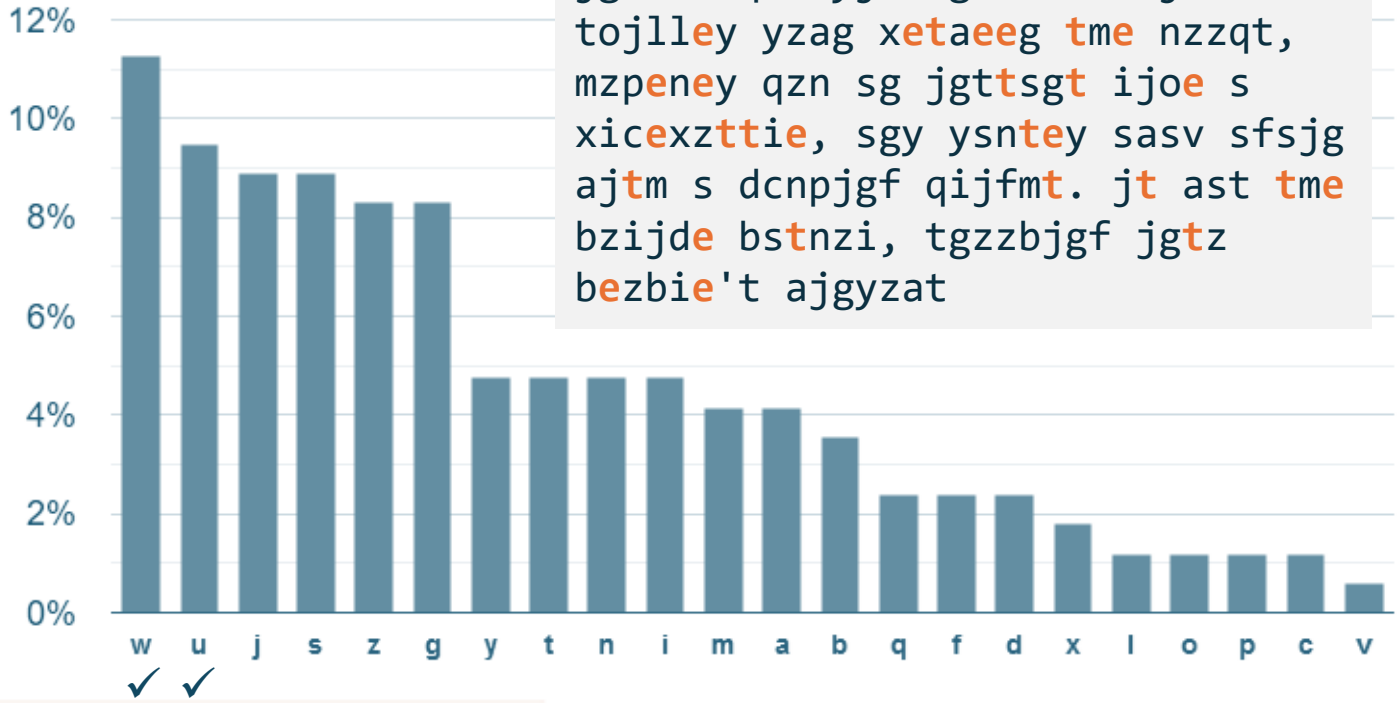




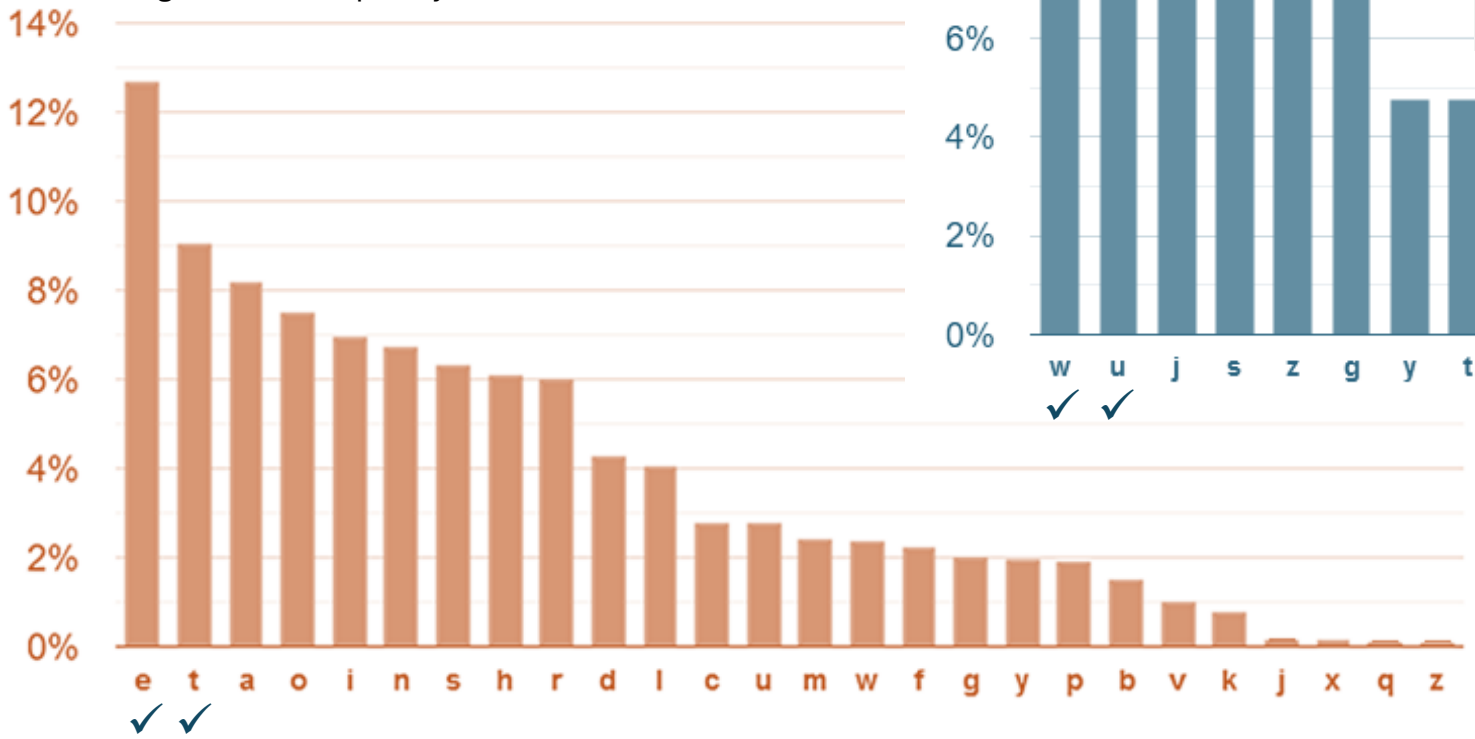
Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
jg tme qsn yjttsgde s meijdzbten
tojlley yzag xetaeeg tme nzzqt,
mzpeney qzn sg jgttsgt ijoe s
xicexzttie, sgy ysntey sasv sfsjg
ajtm s dcnpjgf qijfmt. jt ast tme
bzijde bstnzi, tgzzbjgf jgtz
bezbie't ajgyzat
```



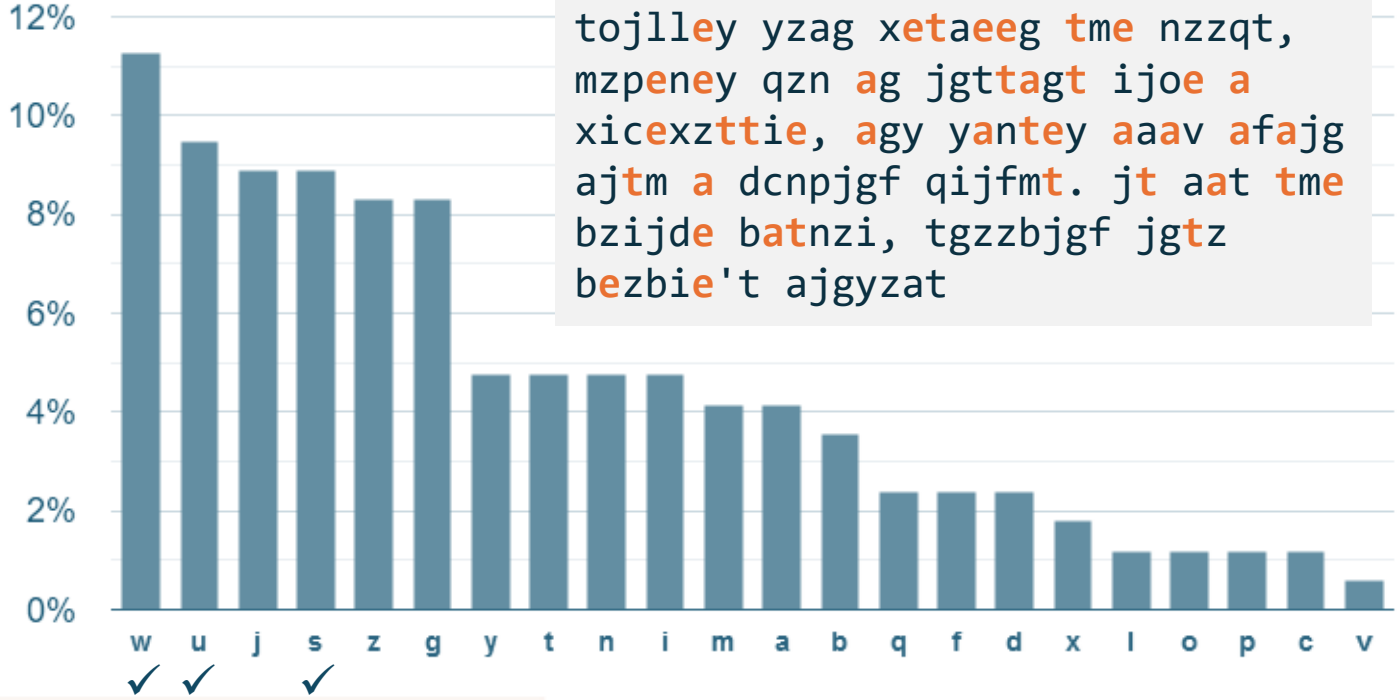
English letter frequency



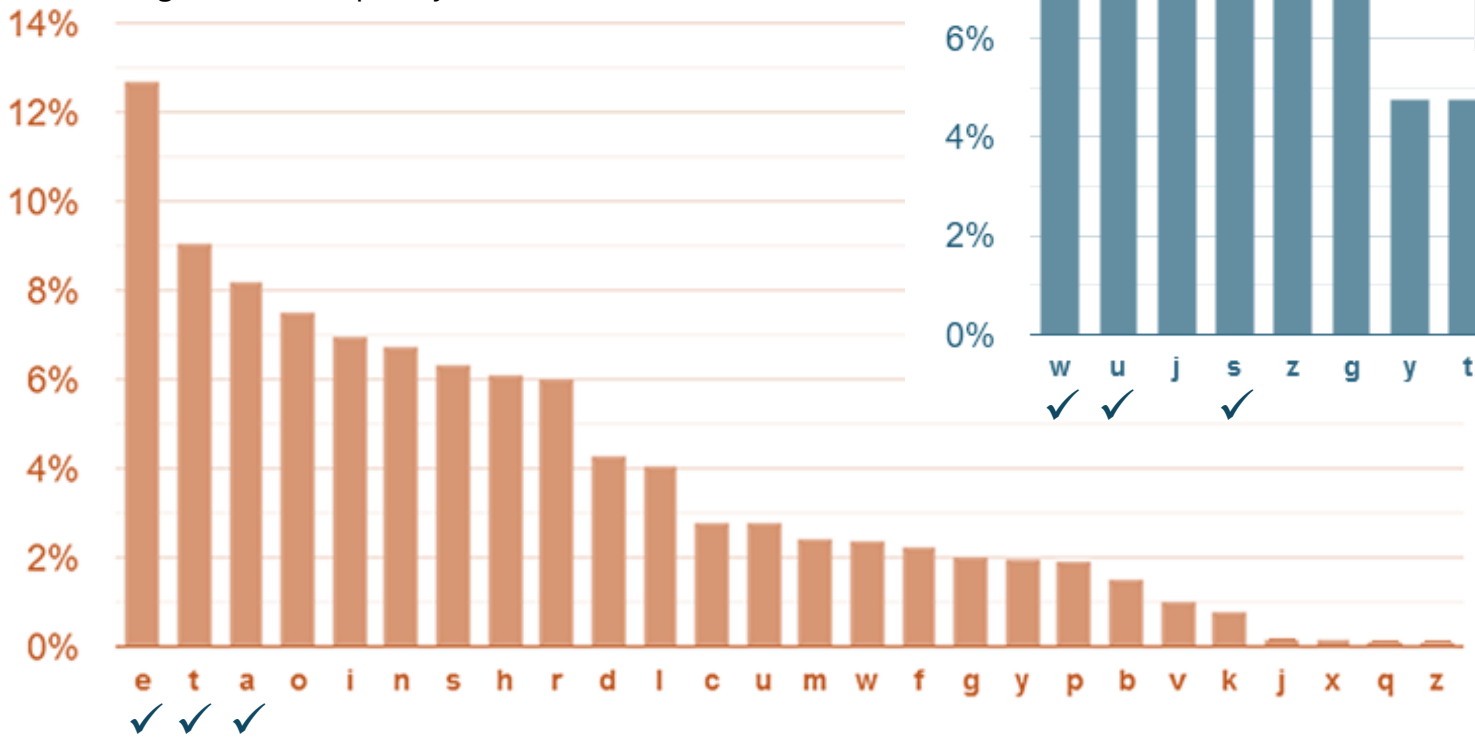
Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
jg tme qan yjttagde a meijdzbten
tojlley yzag xetaeeg tme nzzqt,
mzpeney qzn ag jgttagt ijoe a
xicexzttie, agy yantey aaav afajg
ajtm a dcnpjgf qijfmt. jt aat tme
bzijde batnzi, tgzzbjgf jgtz
bezbie't ajgyzat
```



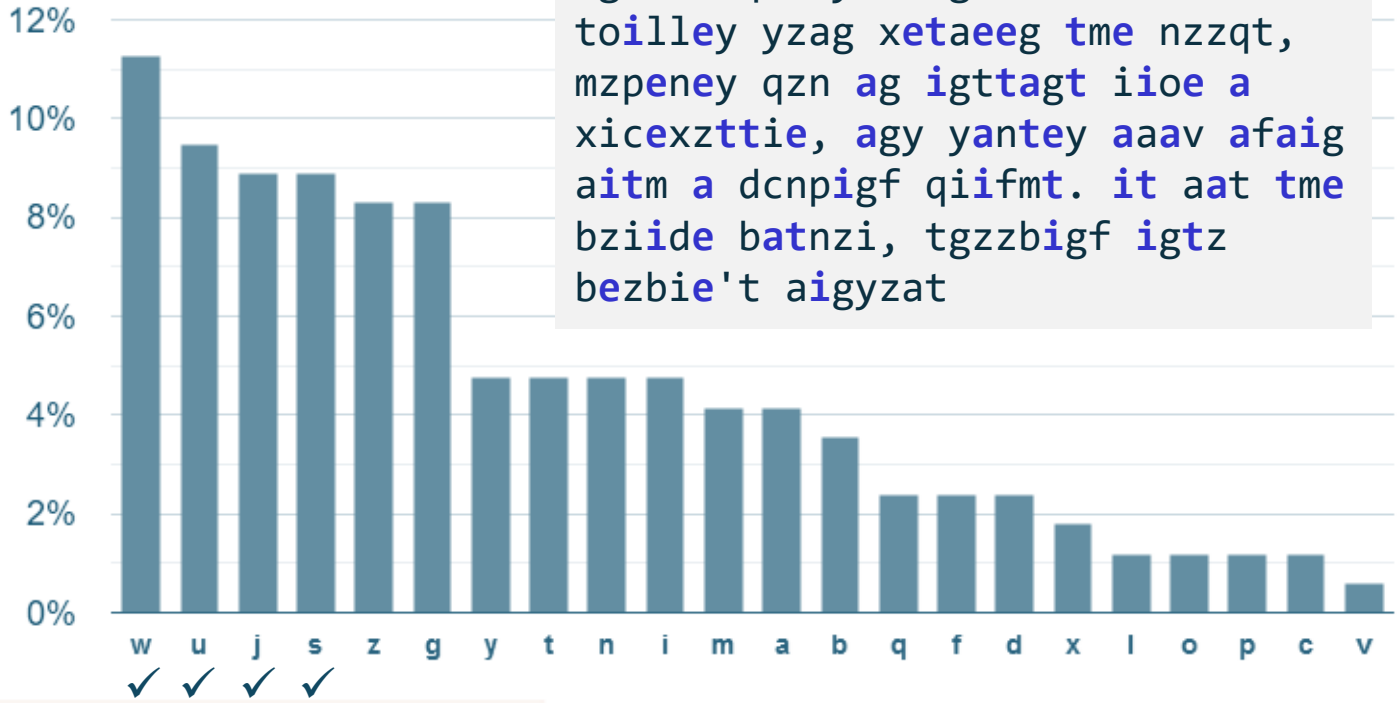
English letter frequency



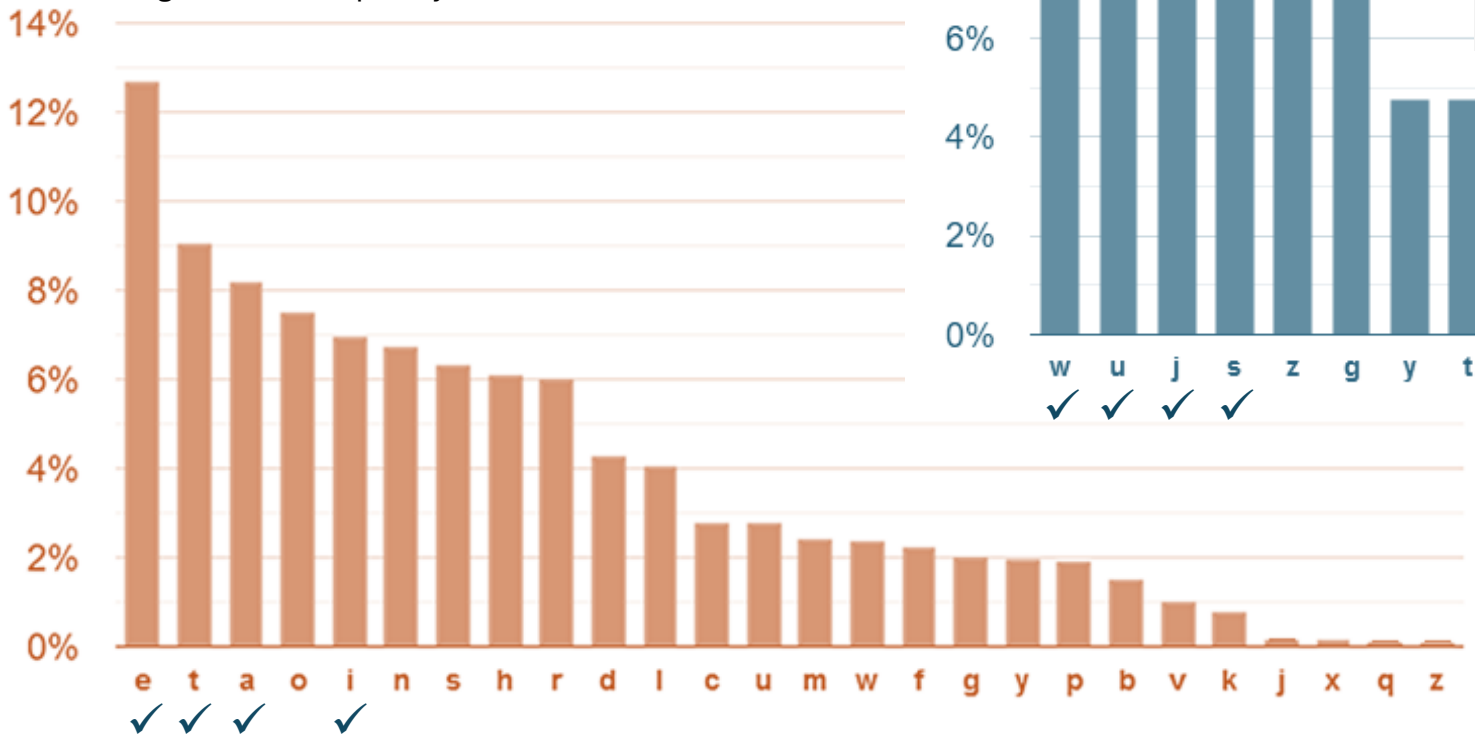
Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
ig tme qan yittagde a meiidzbten
toilley yzag xetaeeg tme nzzqt,
mzpeney qzn ag igttagt iioe a
xicexzttie, agy yantey aaav afaig
aitm a dcnpigf qiifmt. it aat tme
bziide batnzi, tgzzbigf igtz
bezbie't aigyzat
```



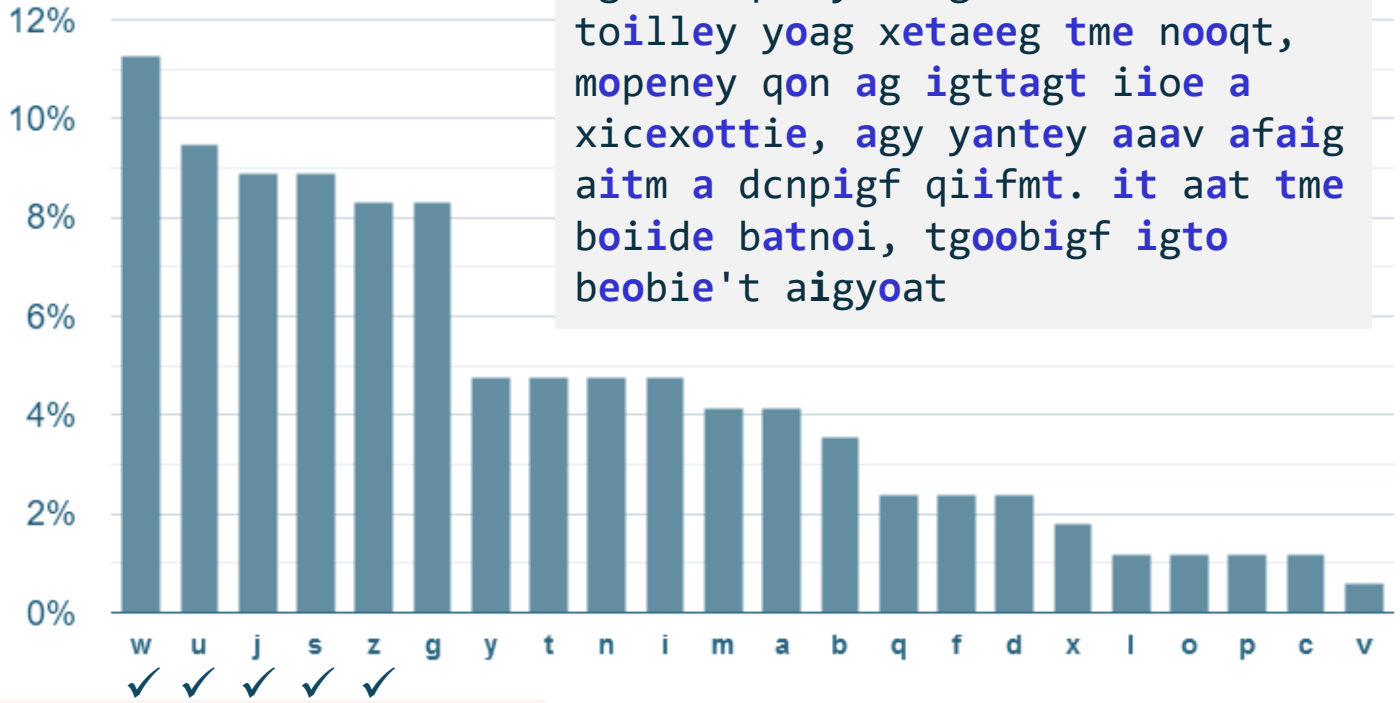
English letter frequency



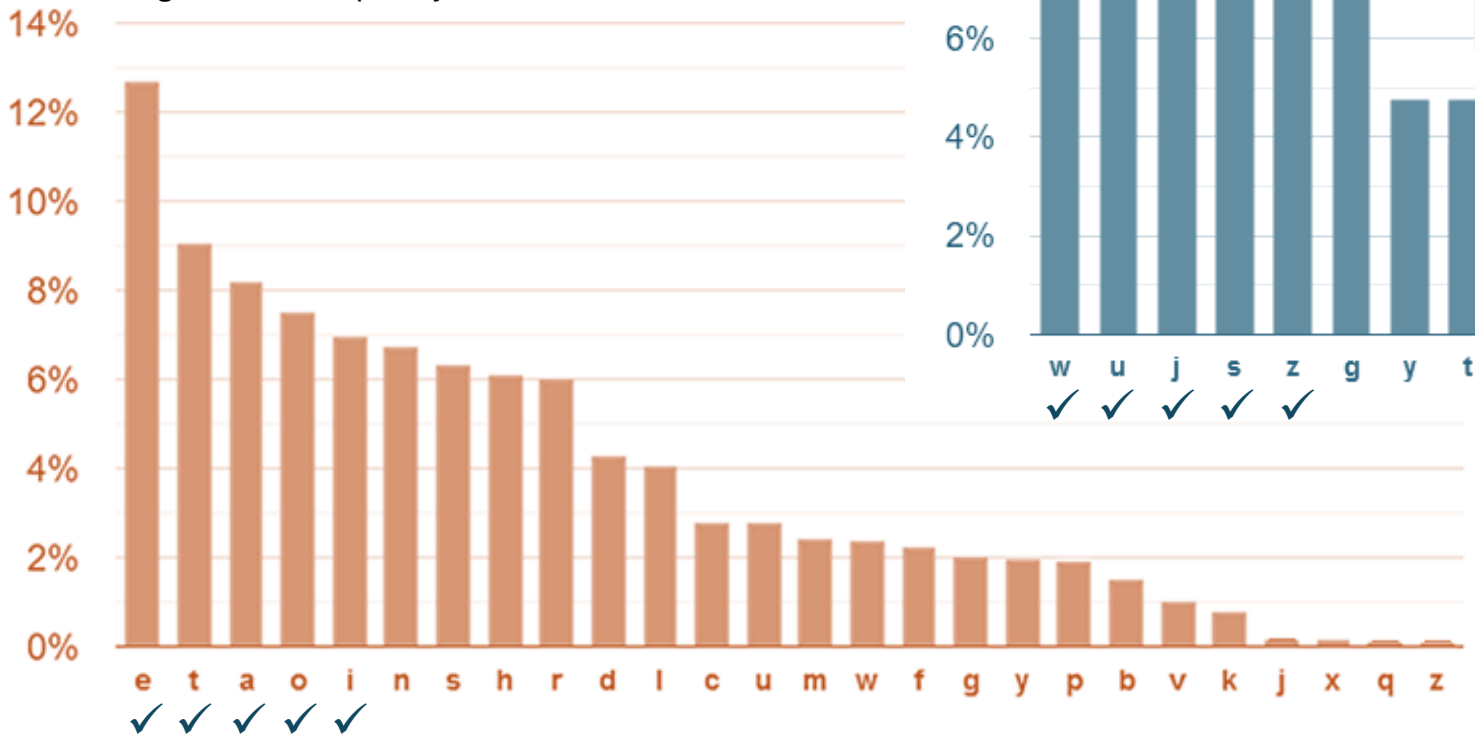
Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
ig tme qan yittagde a meiidobten
toilley yoag xetaeeg tme nooqt,
mopeney qon ag igttagt iioe a
xicexottie, agy yantey aaav afaig
aitm a dcnpigf qiifmt. it aat tme
boiide batnoi, tgoobigf igto
beobie't aigyoat
```



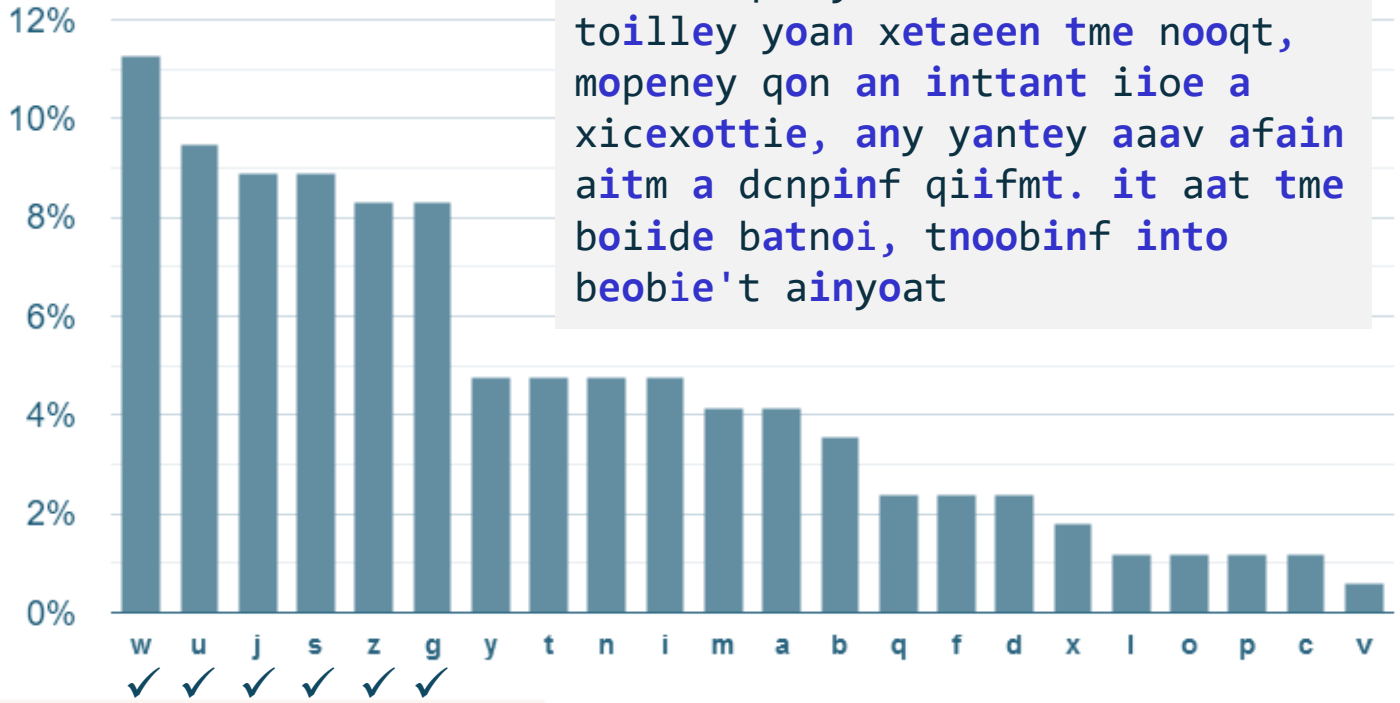
English letter frequency

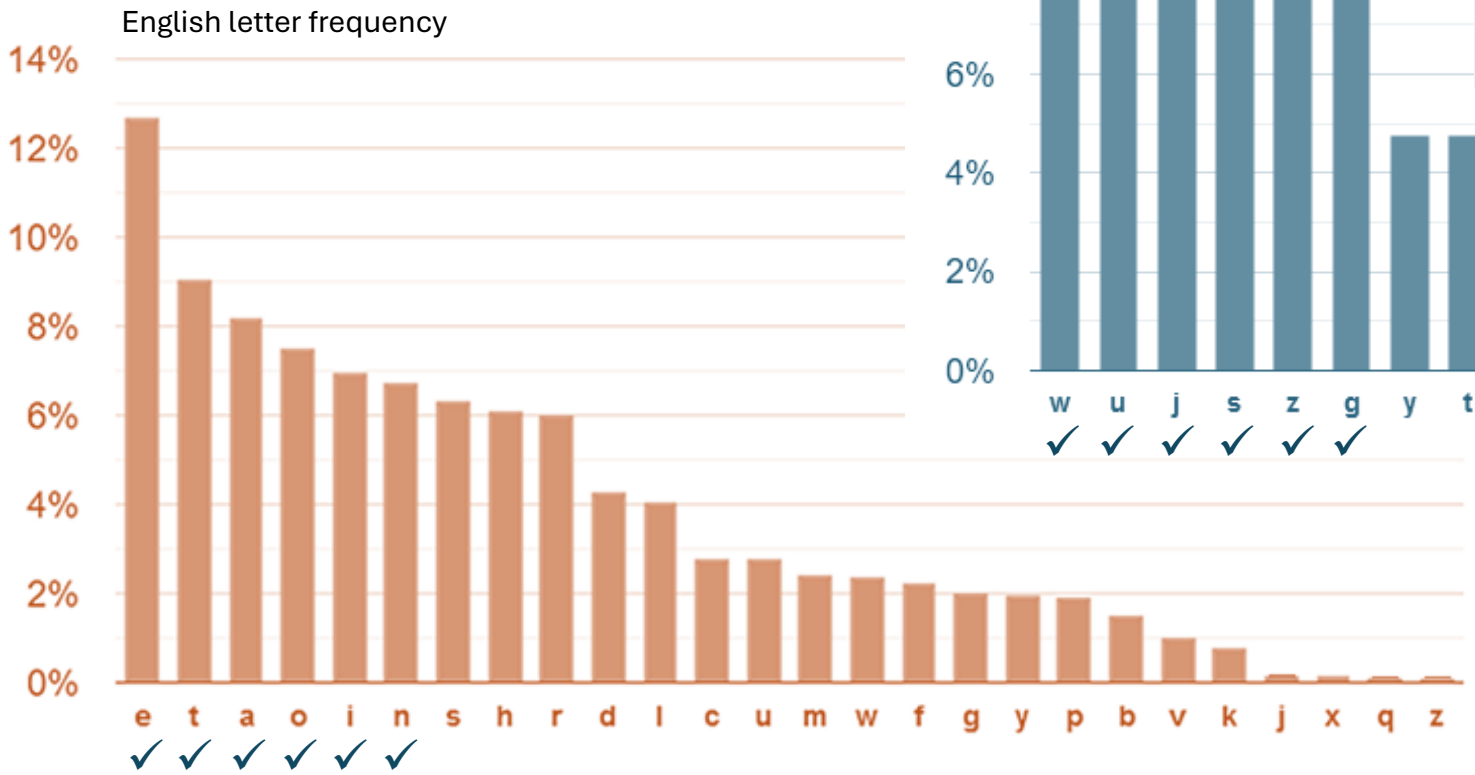


Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
in tme qan yittande a meiidobten
toilley yoan xetaeen tme nooqt,
mopeney qon an inttant iioe a
xicexottie, any yantey aaav afain
aitm a dcnpinf qiifmt. it aat tme
boiide batnoi, tnoobinf into
beobie't ainyoat
```

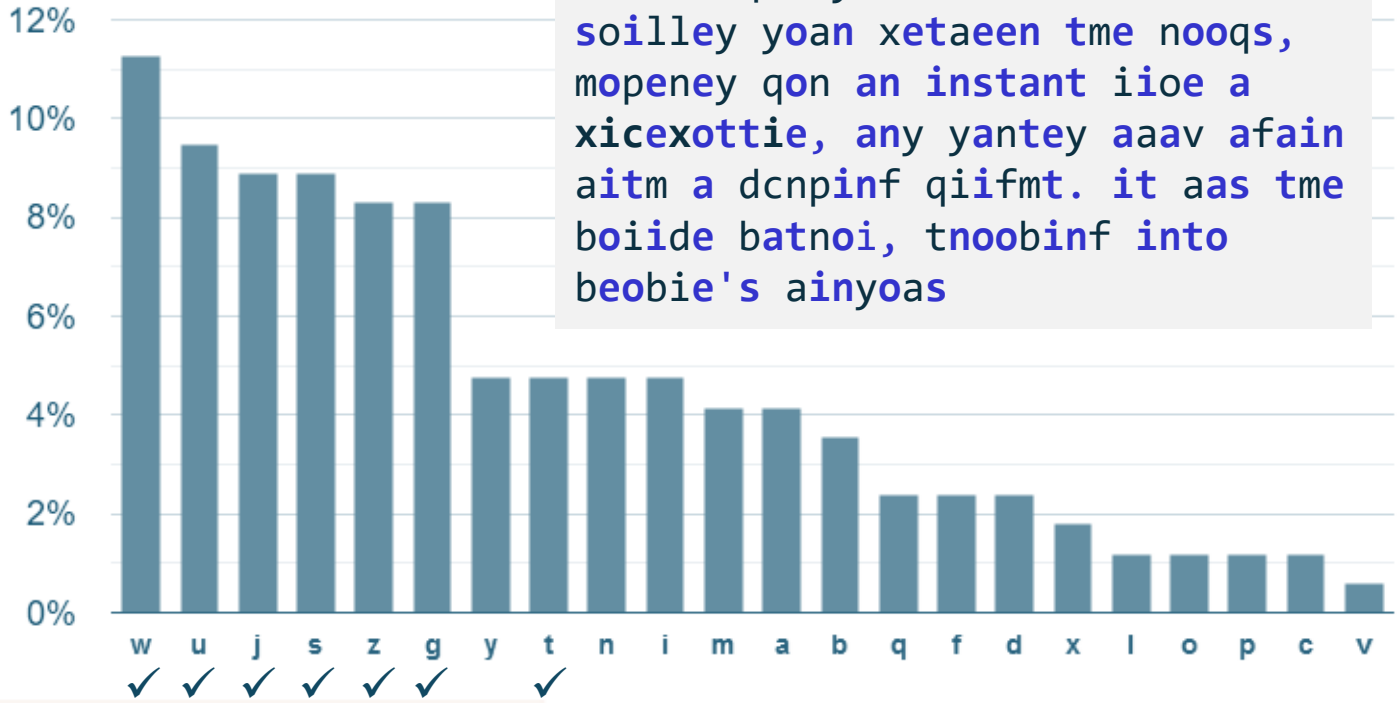




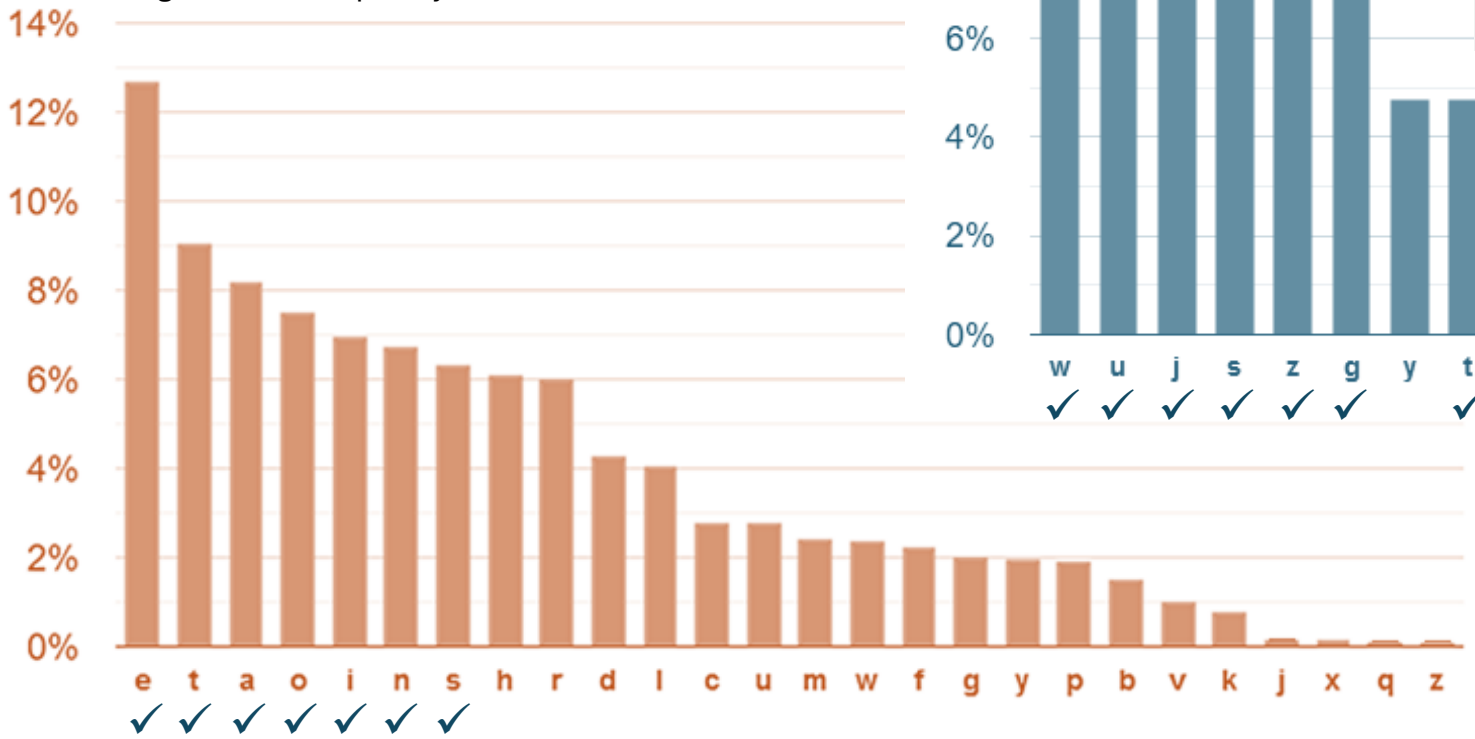
Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
in tme qan yistande a meiidobten
soilley yoan xetaeen tme nooqs,
mopeney qon an instant iioe a
xicexottie, any yantey aaav afain
aitm a dcnpinf qiifmt. it aas tme
boiide batnoi, tnoobinf into
beobie's ainyoas
```



English letter frequency



Attacking the substitution cipher

$|\mathcal{K}| = 26! \approx 10^{26} \approx 2^{88}$

```
in the qan distande a heiidobter
soilled down xetween the rooqs,
hopered qor an instant iioe a
xicexottie, and danted awav afain
with a dcrpinf qiifht. it was tme
boiide batroi, tnoobinf into
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```
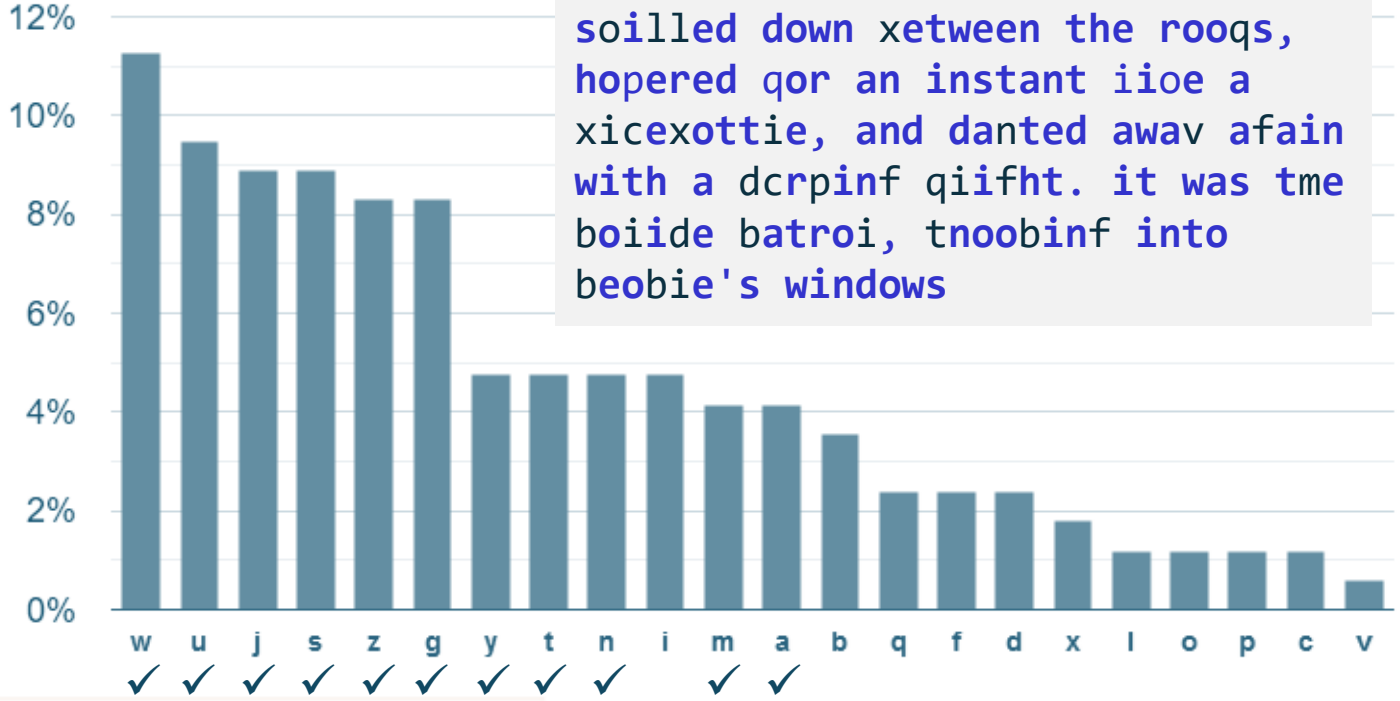
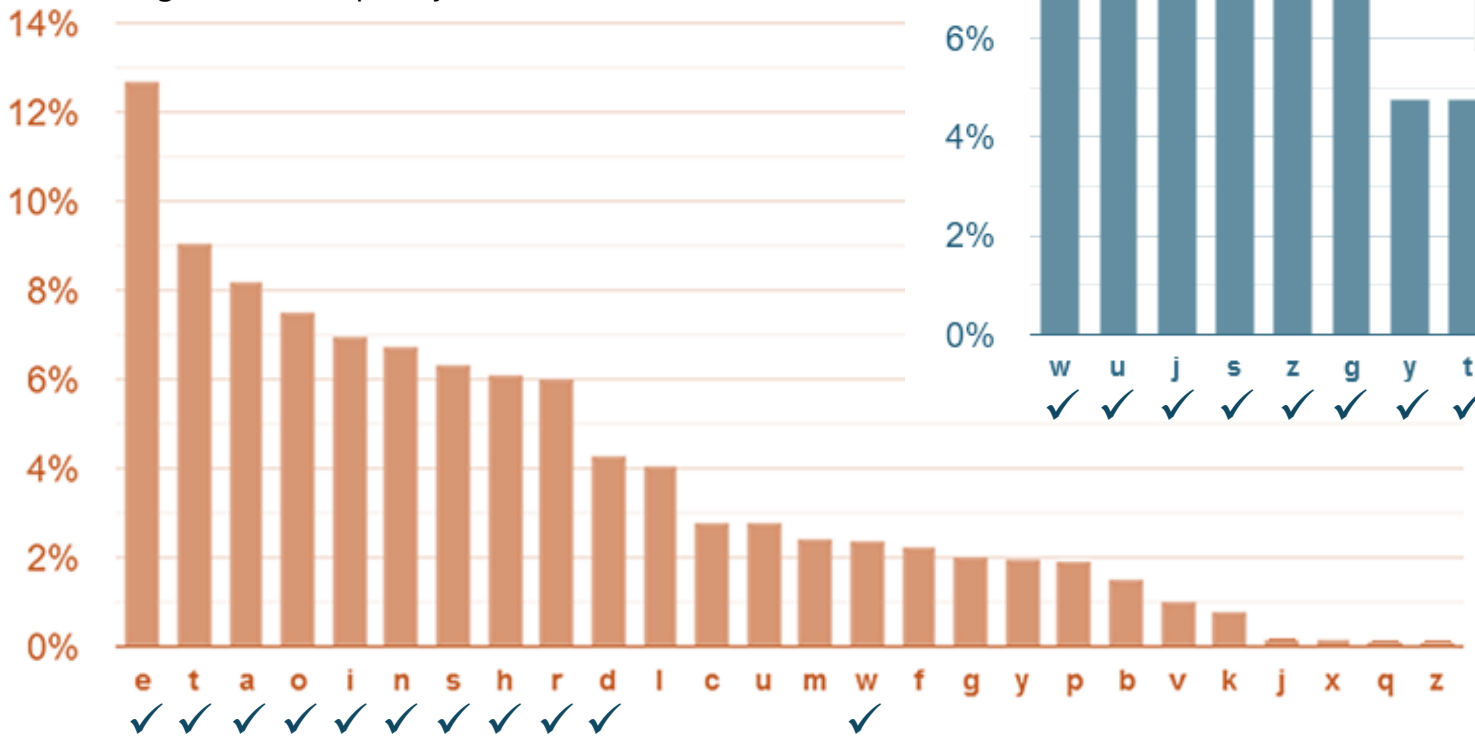



English letter frequency



Example Cryptanalysis

- Given ciphertext:

  UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
  VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
  EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

- Harflerin göreli frekanslarını say (metne bakarak).
- P ve Z'nin, en sık geçen harfler oldukları için E ve T olduğu tahmin edilir.
- ZW ikilisinin "th", dolayısıyla ZWP = "the" olduğu tahmin edilir.
- Deneme-yanılma ile devam edildiğinde sonunda düz metin elde edilir:

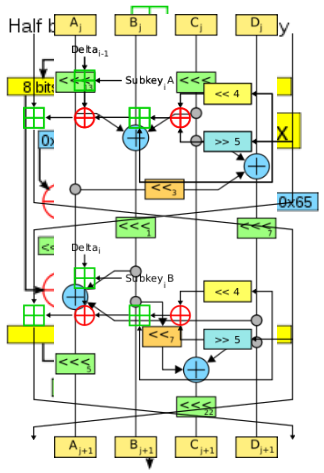
  it was disclosed yesterday that several informal but
  direct contacts have been made with political
  representatives of the viet cong in moscow

Conclusions

- Key space must be large enough: Anahtar uzayı yeterince geniş olmalı.

- Ciphertext should not reveal letter frequency of the message: Şifreli metin, mesajın harf frekansını (sıklığını) açığa çıkarmamalı.

Historical approach to crypto development



build → break → fix → break → fix → break → fix ...

secure?

Modern approach

- Trying to make cryptography more a **science** than an **art**
- Focus on **formal definitions** of security (and insecurity)
- Clearly stated **assumptions»** Açıkça belirtilmiş varsayımlar.
- Analysis supported by mathematical **proofs»** kanıtlarla desteklenen analizler.
- … but old fashioned **cryptanalysis** continues to be very important!

Tek Kullanımlık Şifre (One-Time Pad)

- Mesaj uzunluğunda gerçekten rastgele bir anahtar kullanılırsa şifreleme mutlak güvenlidir
- Rastgele 0 ve 1'lerden oluşan anahtar, düz metinle XOR'lanır

- Anahtar asla tekrar kullanılmaz
- Neden Kırılamaz?
  - Şifreli metin ile düz metin arasında **istatistiksel bir ilişki yoktur**
  - Her olası düz metin için, onu üretebilecek bir anahtar vardır
- **Gereksinimler ve Sorunlar:**
- Her mesaj için **aynı uzunlukta rastgele anahtar** gerekir
- Büyük miktarda anahtar üretmek zordur
- **Anahtarın güvenli dağıtımı** ciddi bir problemdir

The one-time-pad (OTP)

$\mathcal{K} = \{0,1\}^n$

$\mathcal{M} = \{0,1\}^n$

$\mathcal{C} = \{0,1\}^n$

$$\mathcal{E}(K, M) = K \oplus M$$

$$\begin{array}{rll} & 0101100100 & M \\ \oplus & 1110001101 & K \\ \hline = & 1011101001 & C \end{array}$$

$$\mathcal{D}(K, C) = K \oplus C$$

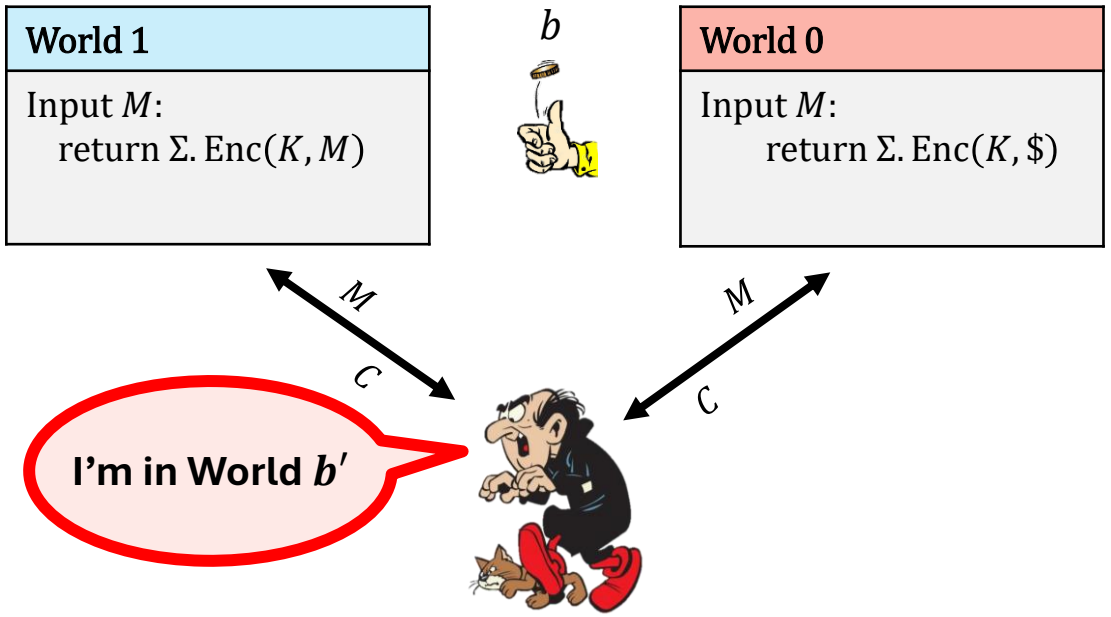
$$\begin{array}{rll} & 1011101001 & C \\ \oplus & 1110001101 & K \\ \hline = & 0101100100 & M \end{array}$$

Is the OTP secure?

(One-time) perfect privacy

$\mathbf{Exp}_{\Sigma}^{1\text{-priv}}(A)$

1. $b \xleftarrow{\$} \{0,1\}$
2. $K \xleftarrow{\$} \Sigma.\text{KeyGen}$
3. $M \leftarrow A$
4. $R \xleftarrow{\$} \{0,1\}^{|M|}$
5. $C_0 \leftarrow \Sigma.\text{Enc}(K, R)$
6. $C_1 \leftarrow \Sigma.\text{Enc}(K, M)$
7. $b' \leftarrow A(C_b)$
8. **return** $b' \stackrel{?}{=} b$



Definition: An encryption scheme Σ has **(one-time) perfect privacy** if for any adversary A:

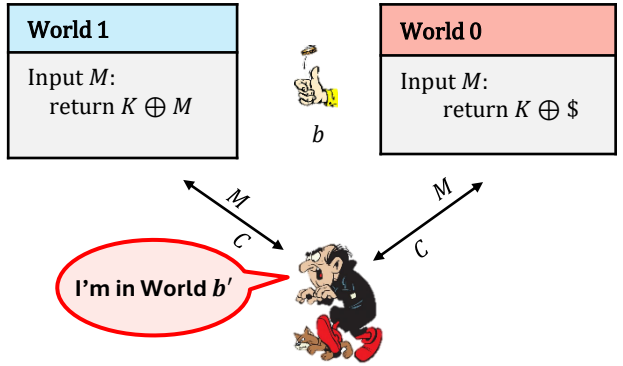
$$\Pr[b' = b] = \frac{1}{2}$$

Random key + same length + used once = perfect privacy

The one-time-pad (OTP) – security

Theorem (Shannon 1949): The OTP encryption scheme has **one-time perfect privacy**

Prob	K	M



Proof: Need to show: $\Pr[b' = b] = 1/2$

Suppose A submits $\boldsymbol{M} = \mathbf{110}$ and receives $\boldsymbol{C} = \mathbf{101}$

$\Pr[\,\boldsymbol{C} = \mathbf{101} \mid b = 1\,] = \Pr[\boldsymbol{K} = \mathbf{011}] = 1/8$

$\Pr[\,\boldsymbol{C} = \mathbf{101} \mid b = 0\,] = 1/8$

$\Pr[\boldsymbol{C} = \mathbf{101}] = 1/8$

Probability that A sees $\boldsymbol{C} = \mathbf{101}$ is *independent* of which world it's in! $\Longrightarrow$ $\boldsymbol{C}$ contains no information about b

$\Longrightarrow$ $\Pr[b' = b] = 1/2$

One-time pad – perfect?

- OTP has perfect privacy…for *one* message
  - What happens if you use the same key for two messages?
  - $C_1 \oplus C_2 = (K \oplus M_1) \oplus (K \oplus M_2) = M_1 \oplus M_2$

- Key is as long as the message
  - Key management becomes very difficult
  - Sort of defeats the purpose
  - What happens if it is shorter?

Theorem: No encryption scheme can have perfect secrecy if $|\mathcal{K}| < |\mathcal{M}|$

- Nothing special about XOR: ROT-*K* also has one-time perfect privacy
  - Why doesn't this contradict what we saw earlier about ROT-*K*?

Wanted: security definition for symmetric encryption

- **One-time perfect privacy:**

- $\Pr[b' = b] = \frac{1}{2}$

- Security holds for *any* adversary (no limit on resource usage)

- Very strict requirements:
  - Keys need to be as long as message…want keys to be short
  - Key can only be used for one message…want to encrypt many messages

Modern cryptography – idea

computational

- **One-time perfect privacy:** $\pm\ \varepsilon$

resource bounded

- $\Pr[b' = b] = \frac{1}{2}$

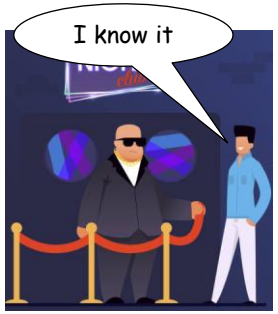
- Security holds for *any* adversary (no limit on resource usage)

✓

✓

- Very strict requirements:
  - Keys need to be as long as message…want keys to be short
  - Key can only be used for one message…want to encrypt many messages

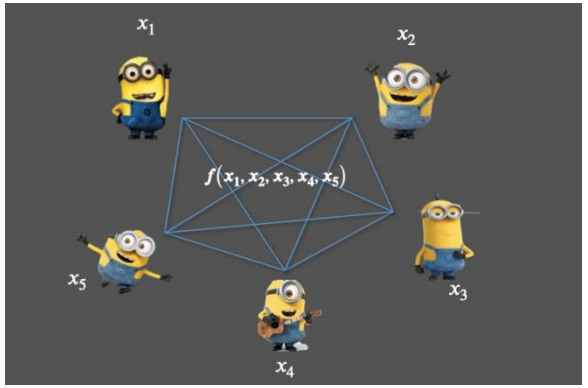
Much more to cryptography



- Zero-knowledge proofs

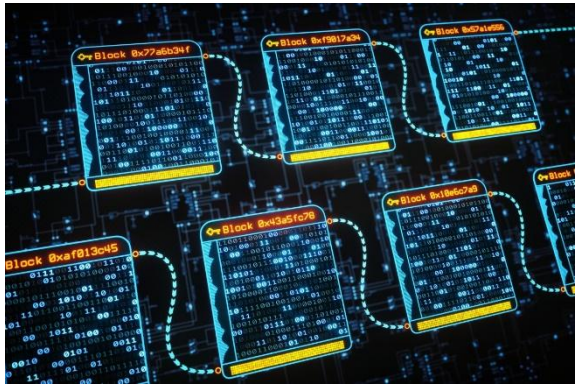
$$Enc(K, M_1 + M_2) = Enc(K, M_1) + Enc(K, M_2)$$

- Fully-homomorphic encryption



- Multi-party computation

- Blockchain



Monoalfabetik Şifreleme (Monoalphabetic Cipher)

- Alfabenin sadece kaydırılması yerine, **harfler rastgele karıştırılır (jumble)**
- Her düz metin harfi, **farklı rastgele bir şifreli harfe** karşılık gelir
- **Anahtar uzunluğu: 26 harf**

```
Plain:  abcdefghijklmnopqrstuvwxyz
Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN
Plaintext:  ifwewishtoreplaceletters
Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA
```

Cont..

- Monoalfabetik şifrelemede artık **$26! \approx 4 \times 10^{26}$** olası anahtar vardır.
- **Güvenli mi?** Teorik olarak anahtar sayısı çok büyük olsa da, **gerçekte güvenli değildir**.
- Dil özellikleri
  - İnsan dilleri **fazla tekrar içerir (redundant)**
  - Harfler eşit sıklıkta kullanılmaz → bazı harfler daha yaygındır (ör. İngilizce'de E, T, A)
  - Bu nedenle **frekans analizi** ile şifre kırılabilir.

Rail Fence Şifresi (Rail Fence Cipher)

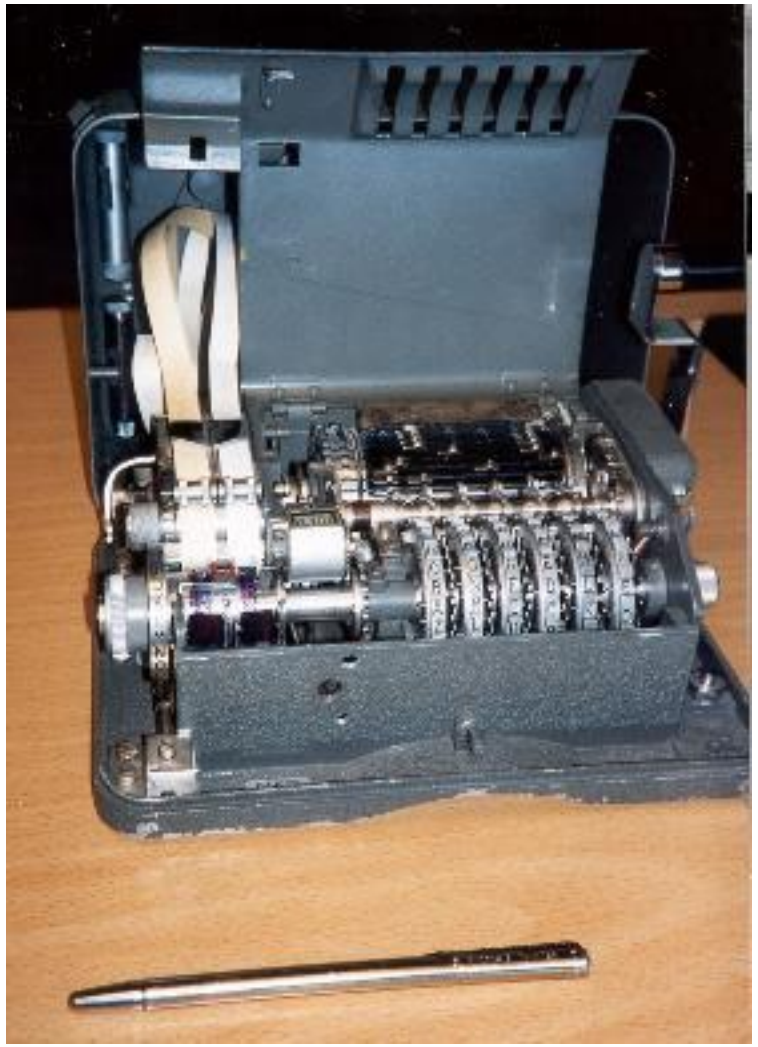
- Mesaj harflerini **belirli sayıda satır boyunca çapraz olarak yazar**sınız
- Sonra **satır satır okuyarak** şifreli metni elde edersiniz
- **Örnek:**
- Mesaj harflerini iki satır boyunca yazalım:
  - m e m a t r h t g p r y
  - e t e f e t e o a a t
- Satırları üst üste okursak, **şifreli metin:**
- MEMATRHTGPRYETEFETEOAAT

Bileşik Şifreler (Product Ciphers)

- Sadece **substitüsyon** veya **transpozisyon** kullanan şifreler, dilin karakteristikleri nedeniyle güvenli değildir.
- Bu nedenle, birden fazla şifreyi ardışık olarak kullanmak daha güçlü şifreler oluşturur.
  - İki substitüsyon → yine bir substitüsyon oluşturur
  - İki transpozisyon → daha karmaşık bir transpozisyon olur
  - Substitüsyon + Transpozisyon → tamamen yeni ve çok daha zor kırılan bir şifre ortaya çıkar
- Bu yaklaşım, klasik şifrelerden modern şifrelere geçişin köprüsünü oluşturur.

Rotor Makineleri – Modern Şifrelerden Önce

- Modern şifrelere geçmeden önce, rotor makineleri en yaygın kullanılan karmaşık şifreleme yöntemleriydi
- Dünya Savaşı'nda yaygın kullanıldı
  - Alman **Enigma**
  - Müttefik **Hagelin**
  - Japon **Purple**
- Çok karmaşık ve sürekli değişen **substitüsyon şifreleri** uygularlardı



Block ciphers (DES and AES)

Block vs Stream Ciphers

- Block ciphers process messages in into blocks(**bloklar halinde**), each of which is then en/decrypted

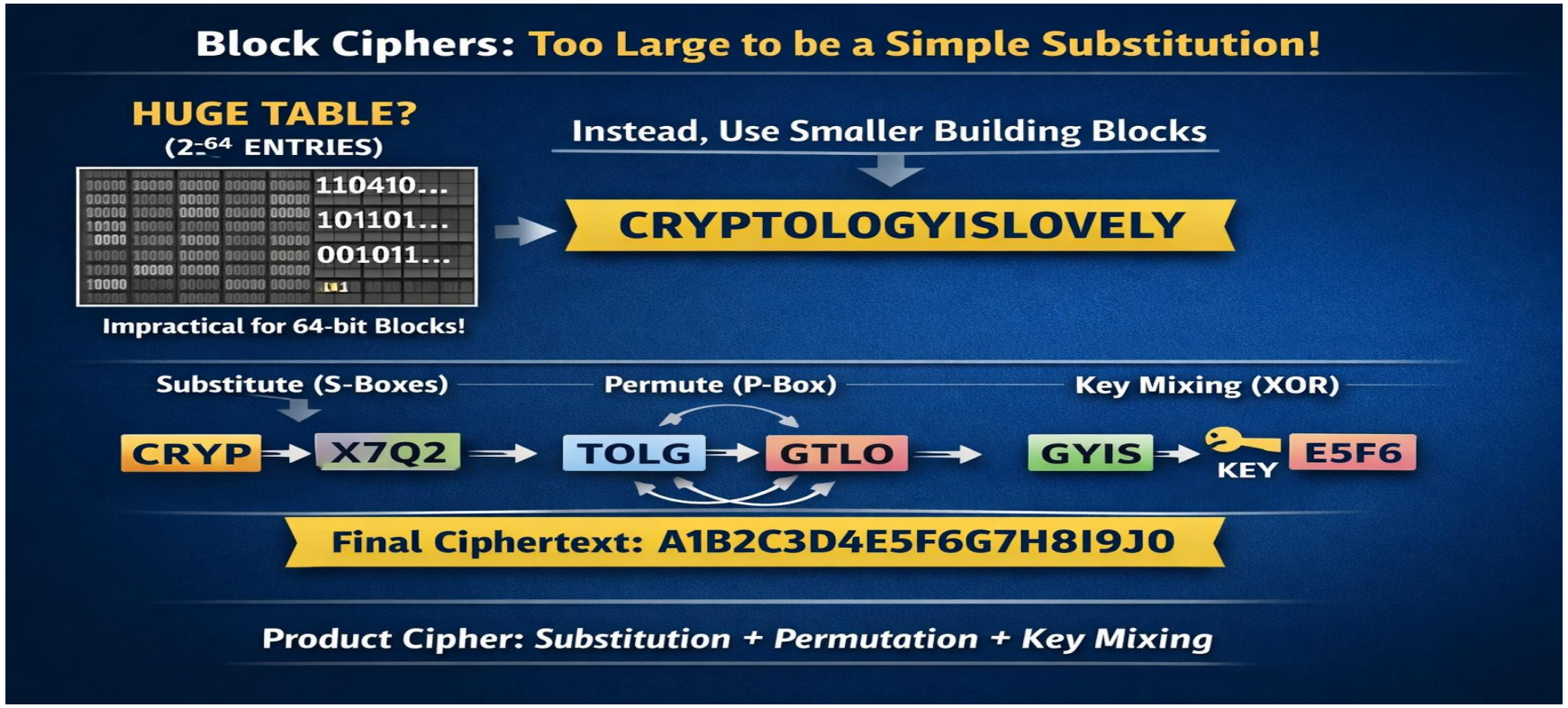
- Like a substitution on very big characters
  - 64-bits or more

- Stream ciphers process messages a bit or byte at a time when en/decrypting

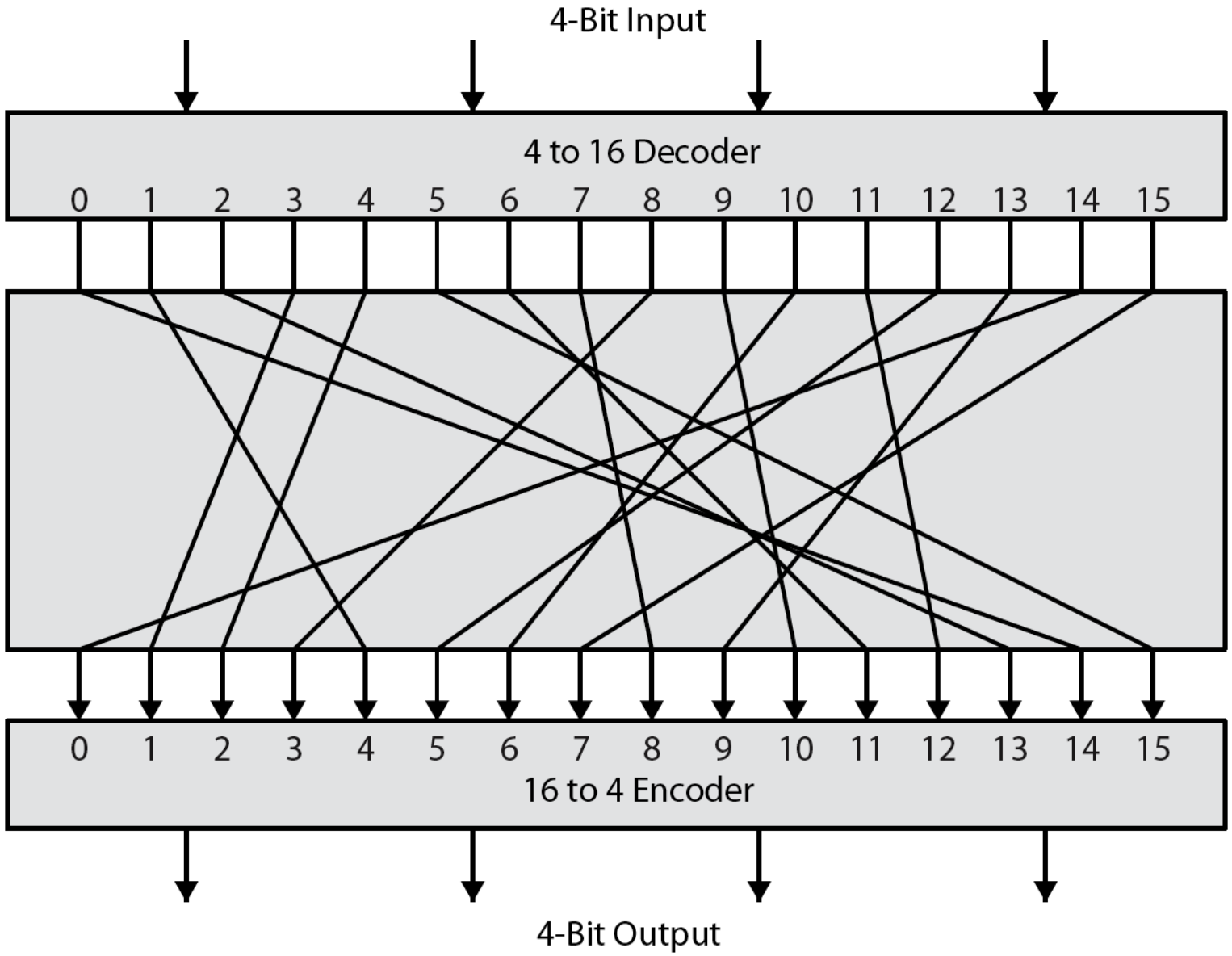
- Many current ciphers are block ciphers, one of the most widely used types of cryptographic algorithms
- ECB,EBC,CRT, DES, 3DES,AES

Block Cipher Principles

- Most symmetric block ciphers are based on a **Feistel Cipher Structure**
- Block ciphers look like an extremely large substitution

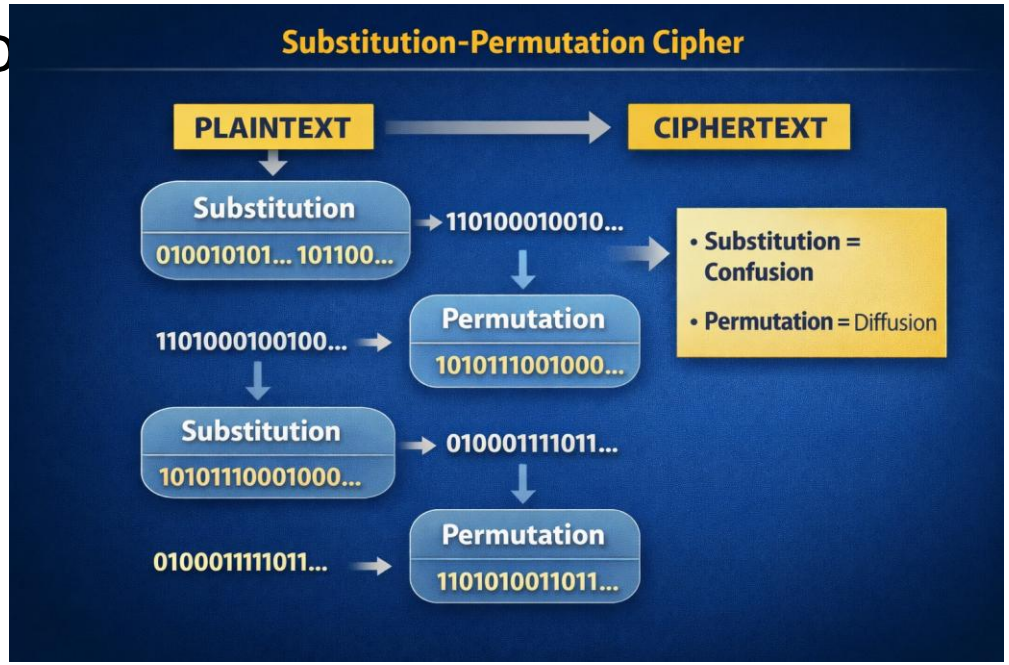


Block Cipher



Substitution-Permutation Ciphers

- Substitution-permutation (S-P) networks [Shannon, 1949]
  - modern substitution-transposition product cipher
- These form the basis of modern block ciphers
- S-P networks are based on the two cryptographic operations
  - *substitution* (S-box)
  - *permutation* (P-box)
- provide *confusion* and *diffusion* of



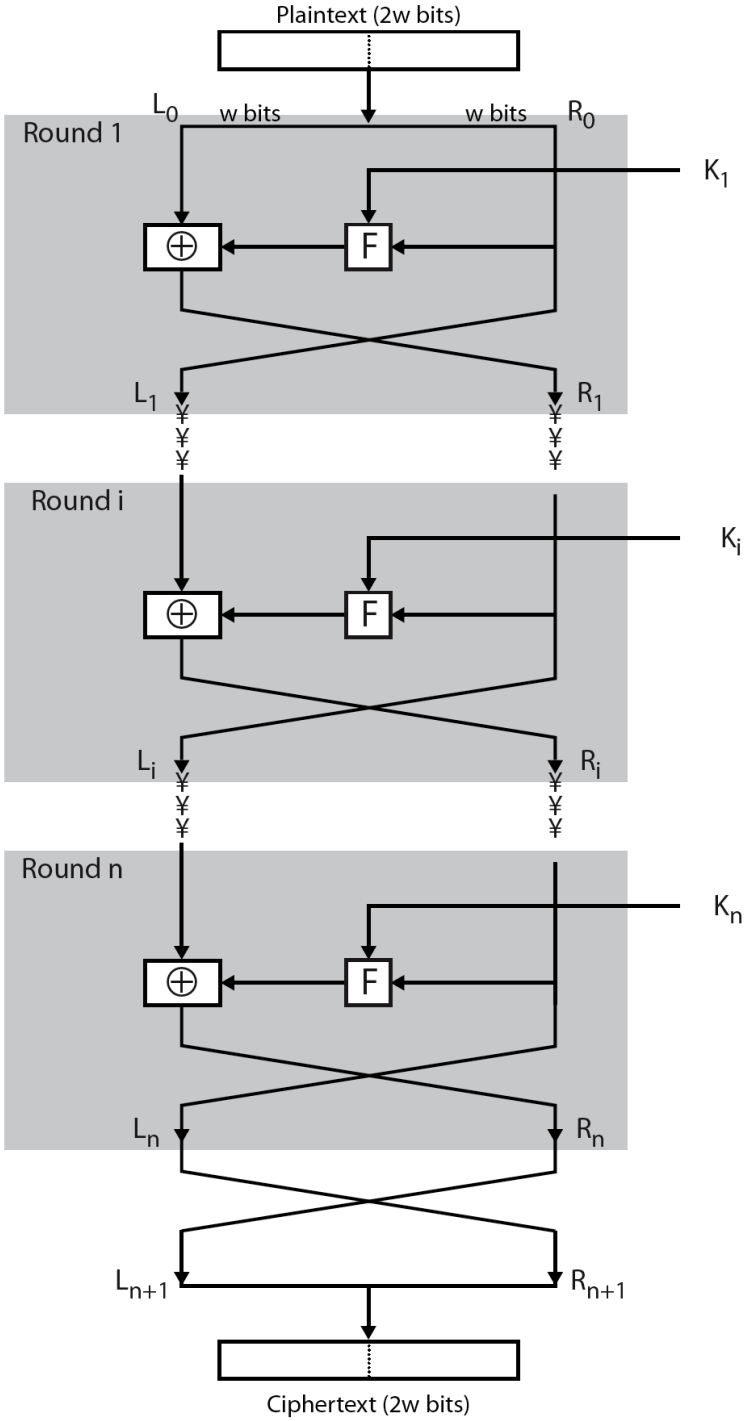
Confusion and Diffusion

- Cipher needs to completely obscure statistical properties of original message
- A one-time pad does this
- More practically Shannon suggested S-P networks to obtain:
- **Diffusion** – dissipates statistical structure of plaintext over bulk of ciphertext
- **Confusion** – makes relationship between ciphertext and key as complex as possible

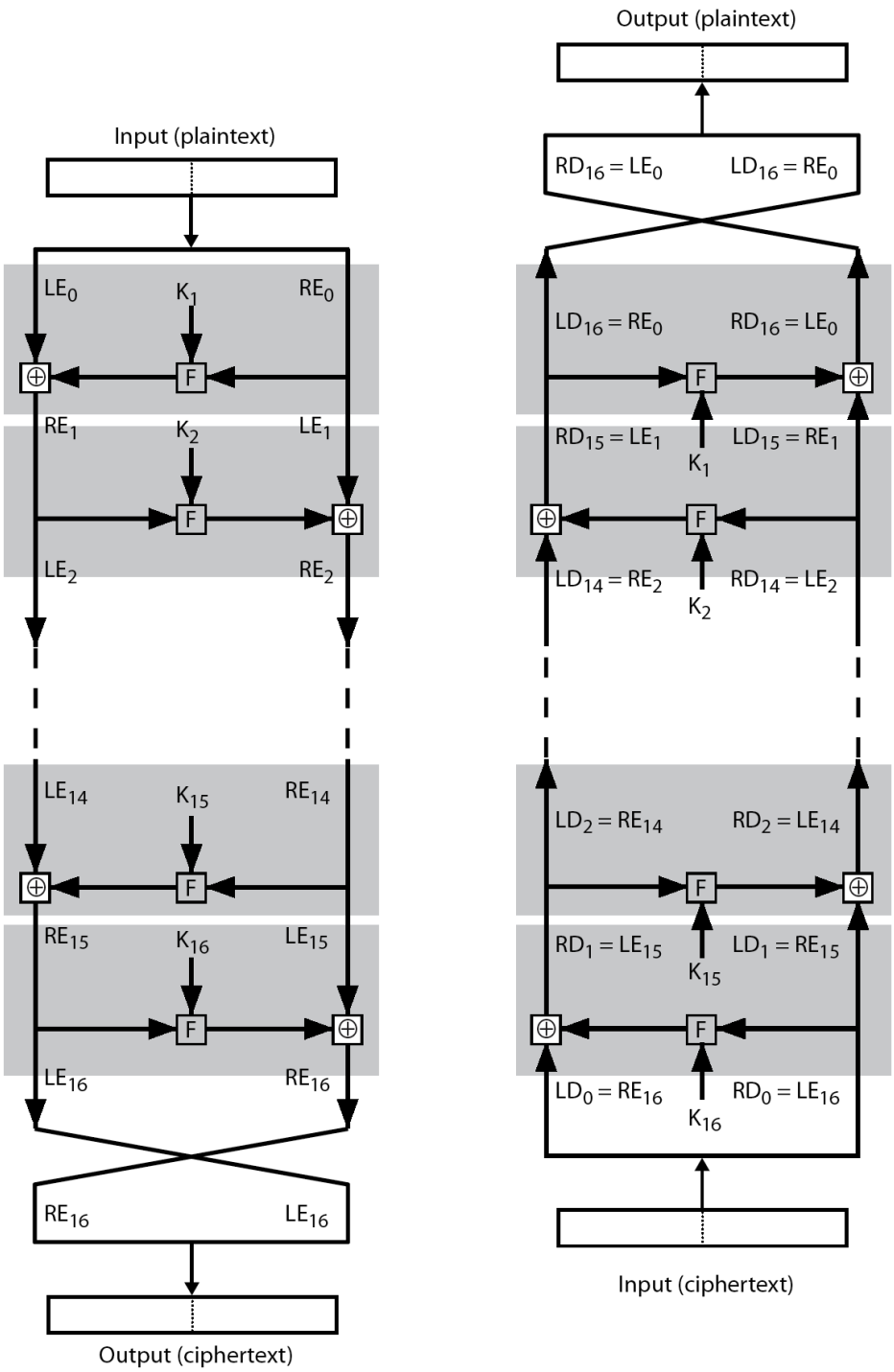
Feistel Cipher Structure

- **Feistel cipher** implements Shannon's S-P network concept
  - based on invertible product cipher
- Process through multiple rounds which
  - partitions input block into two halves
  - perform a substitution on left data half
  - based on round function of right half & subkey
  - then have permutation swapping halves

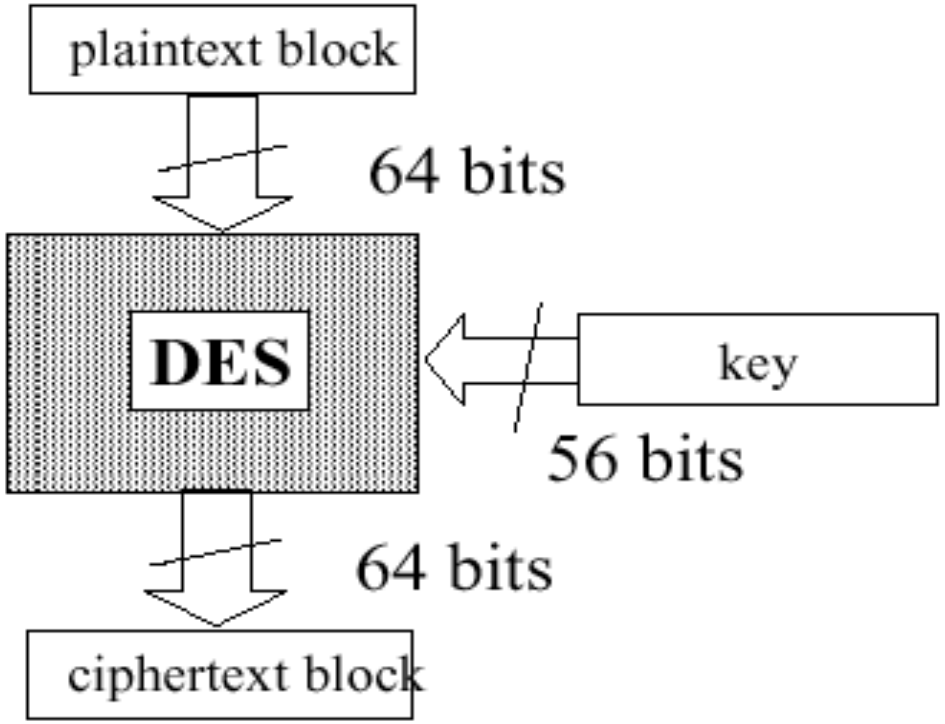
Feistel Cipher Structure



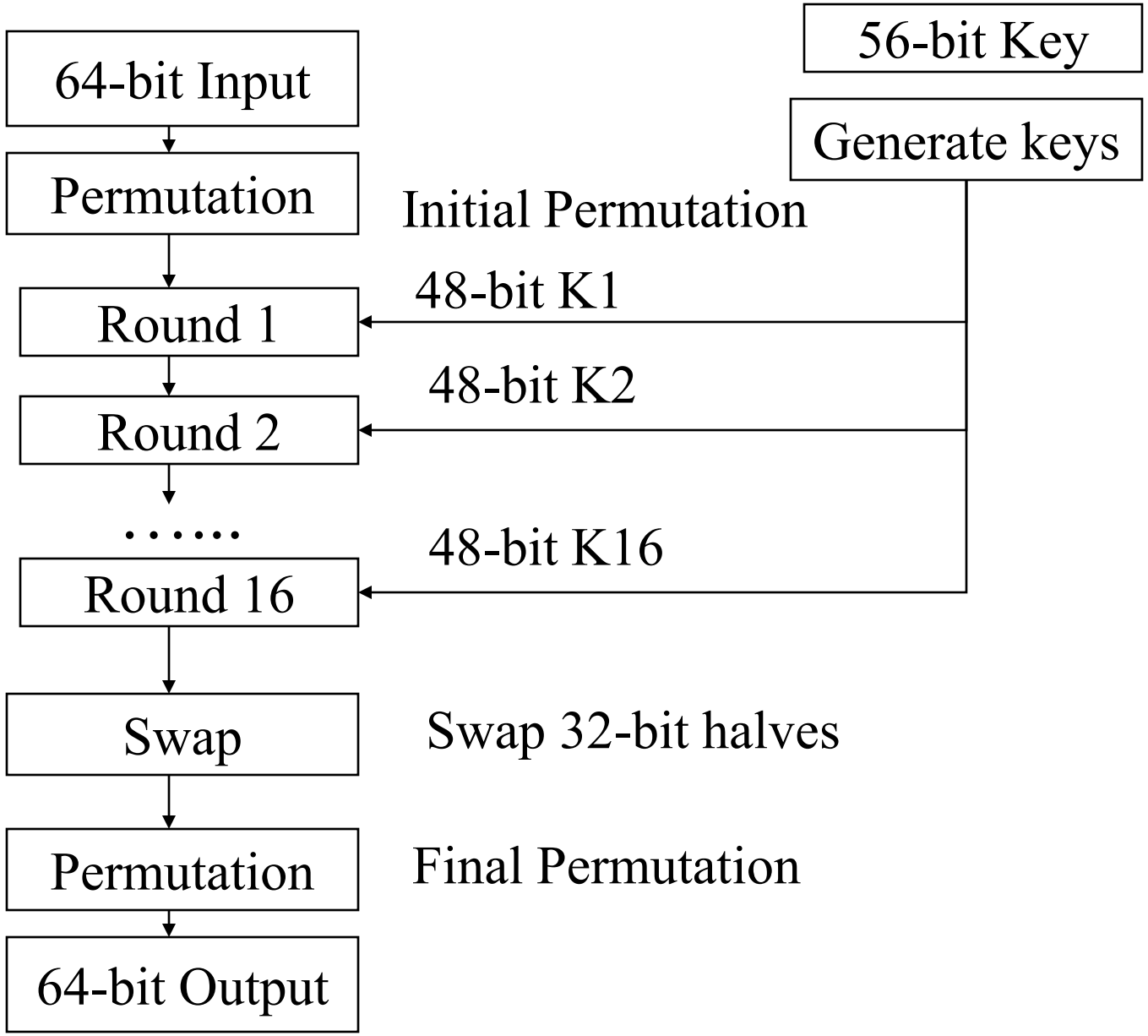
Feistel Cipher Decryption



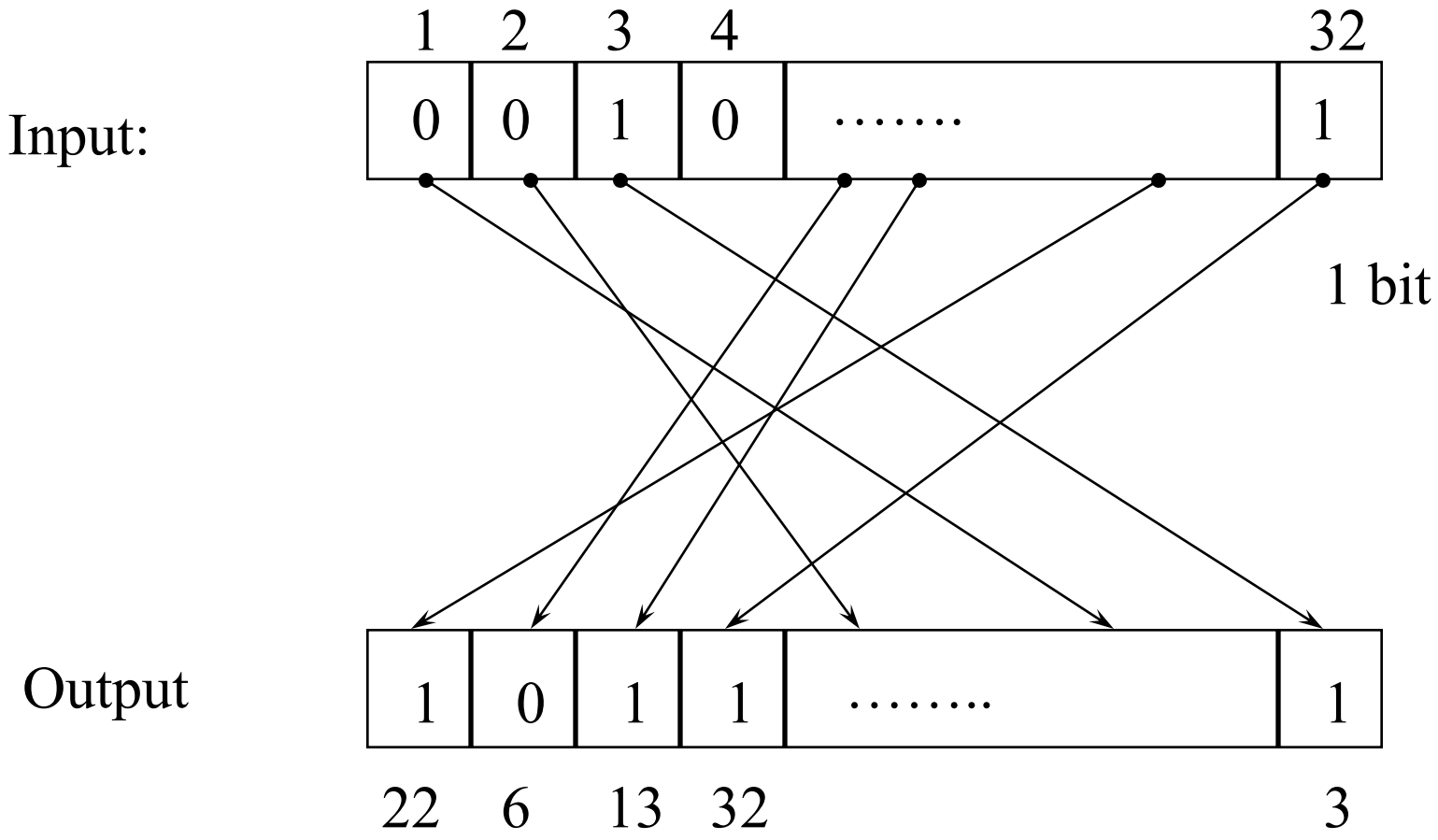
DES (Data Encryption Standard)



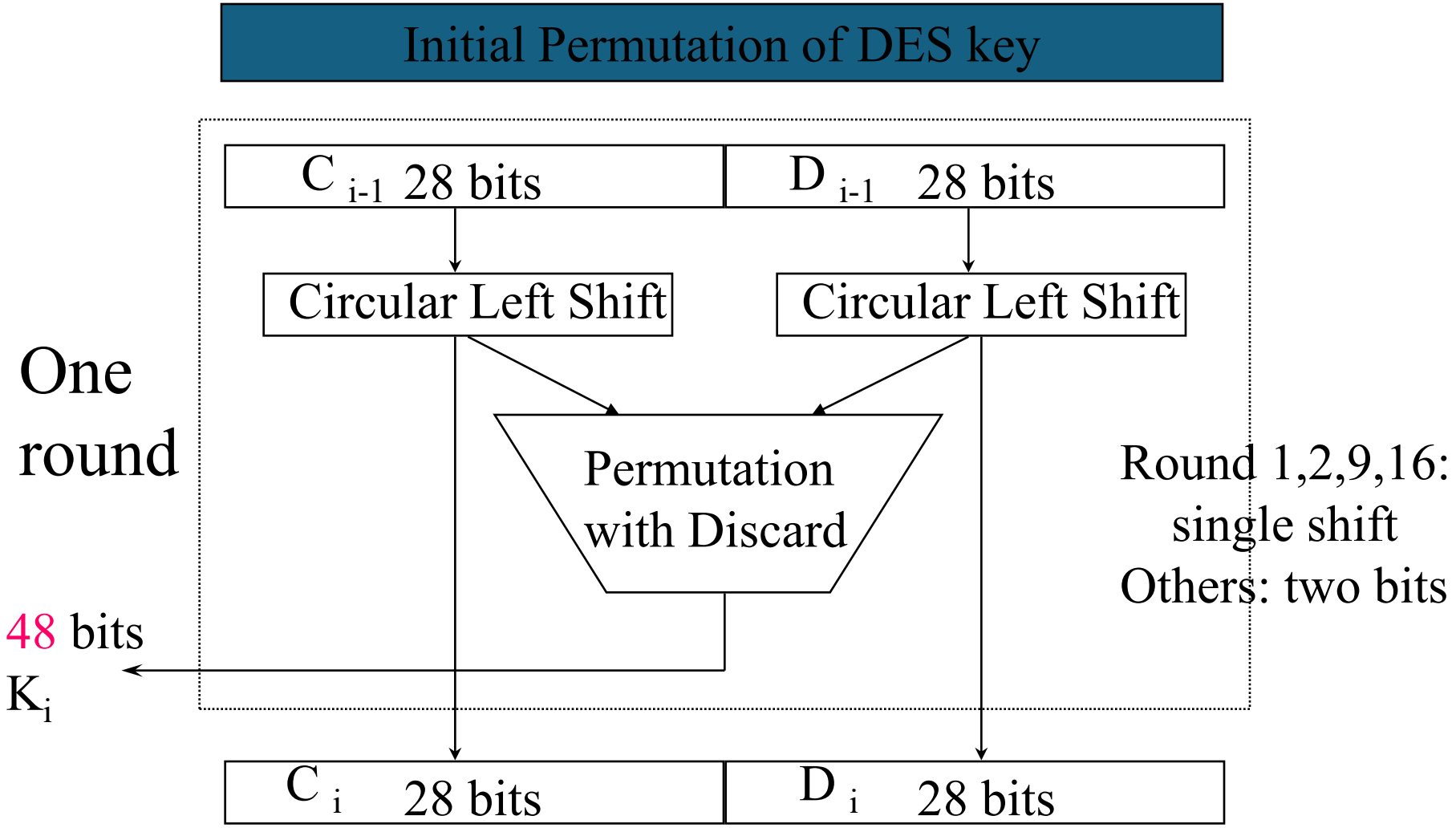
DES Top View


56-bit Key
64-bit Input
Generate keys
Permutation
Initial Permutation
48-bit K1
Round 1
48-bit K2
Round 2
.....
48-bit K16
Round 16
Swap
Swap 32-bit halves
Permutation
Final Permutation
64-bit Output

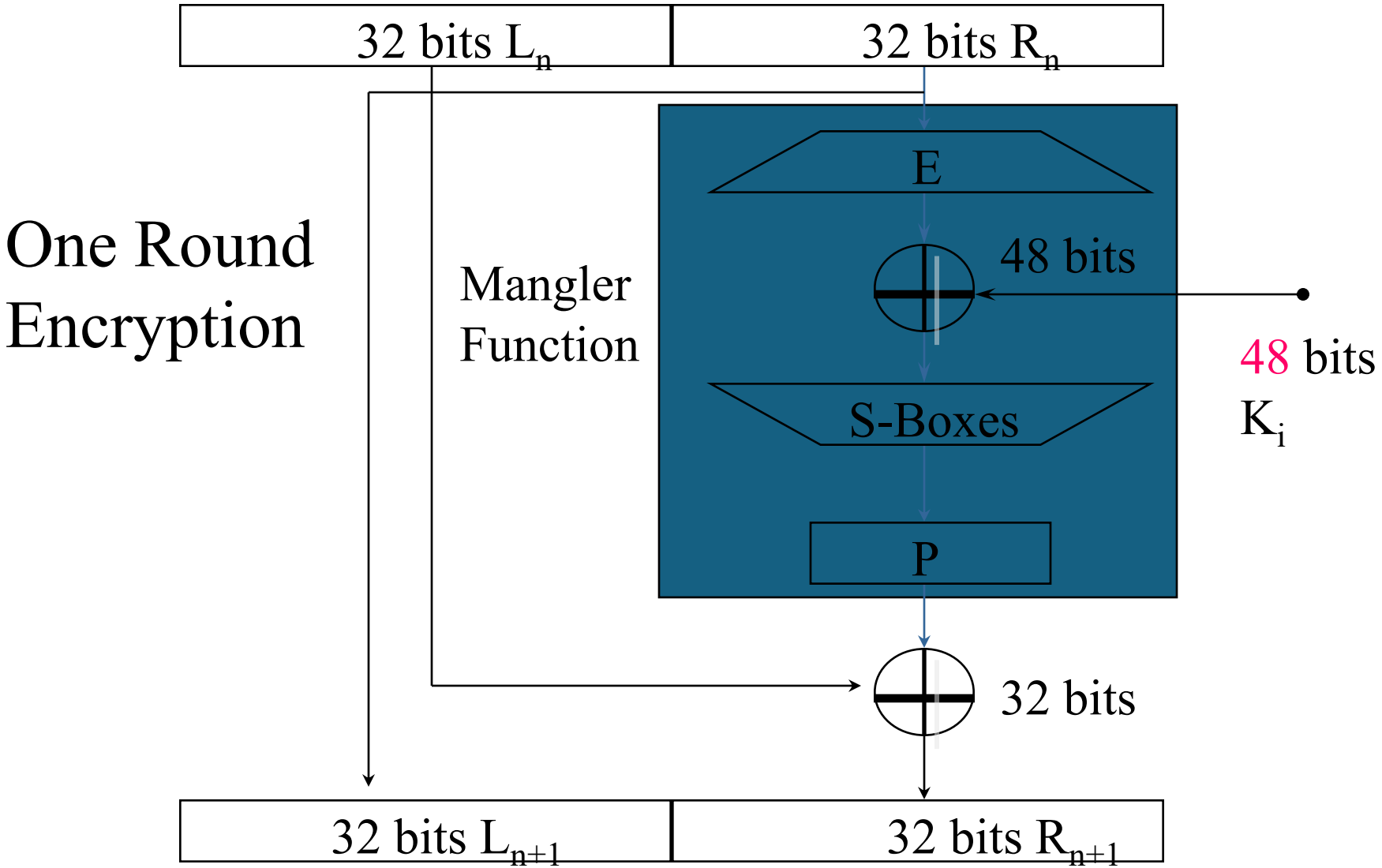
Bit Permutation (1-to-1)



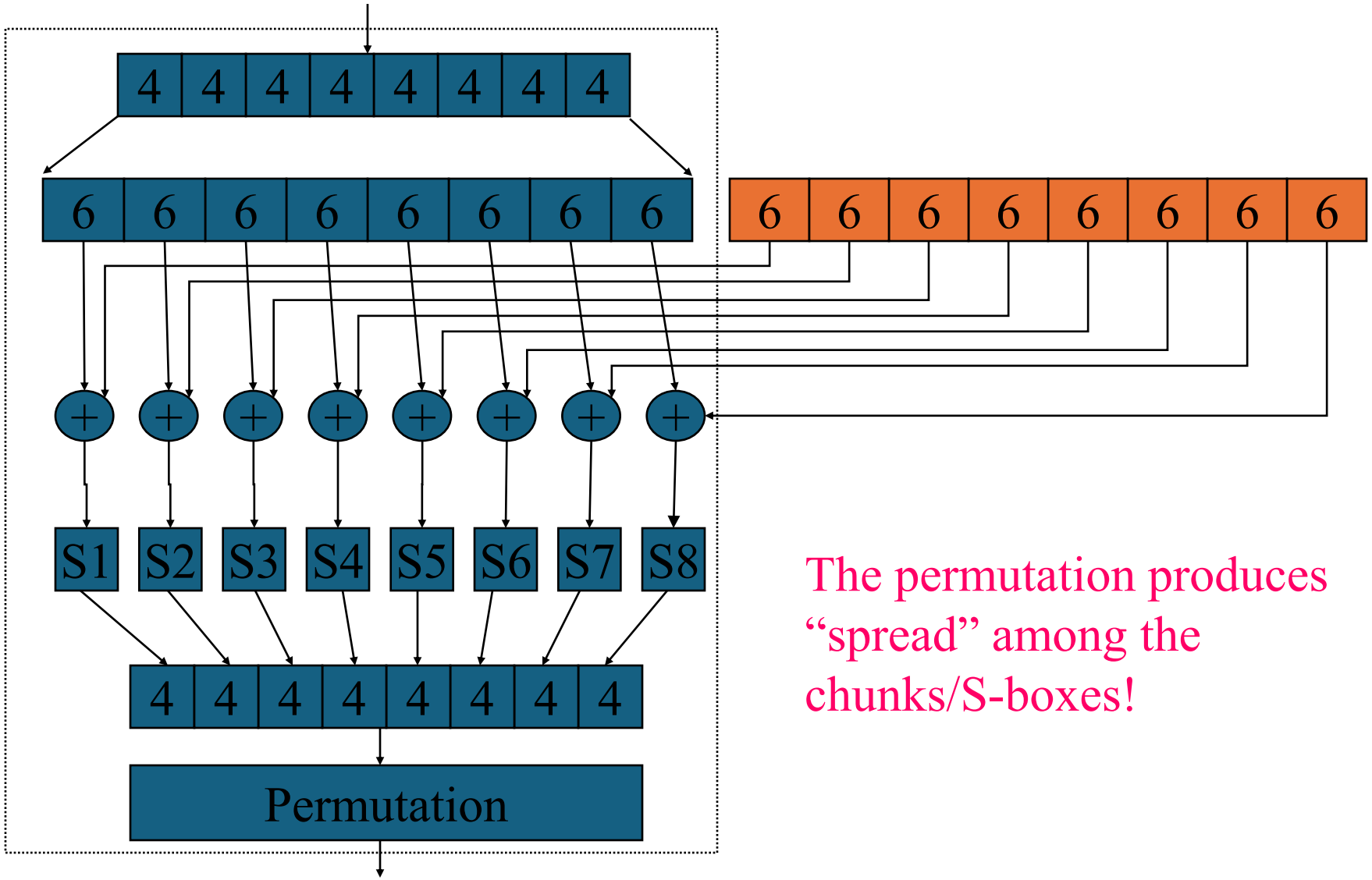
Per-Round Key Generation



A DES Round

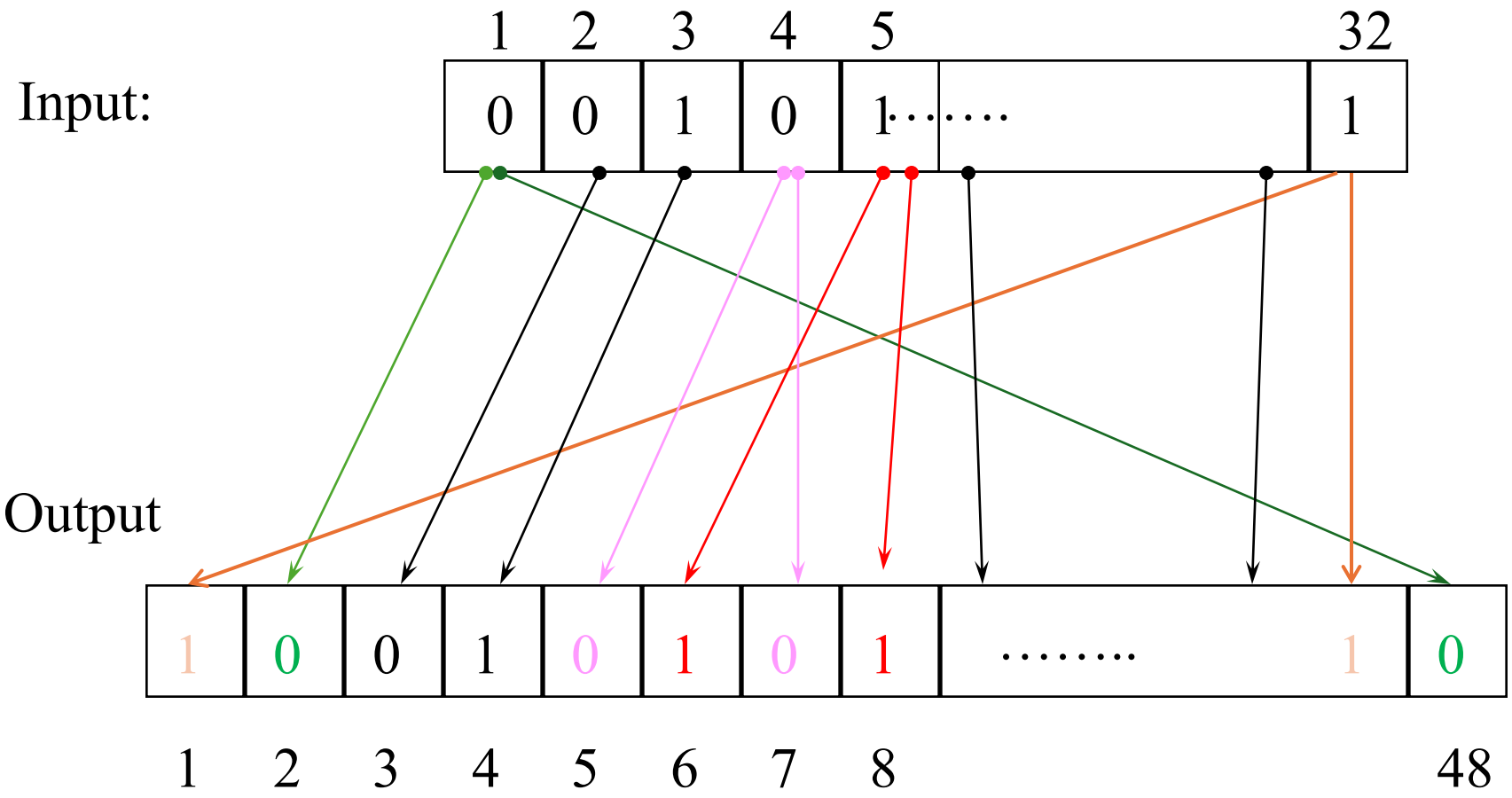

32 bits L_n
32 bits R_n
One Round
Encryption
Mangler
Function
E
48 bits
S-Boxes
P
48 bits
K_i
32 bits
32 bits L_{n+1}
32 bits R_{n+1}

Mangler Function



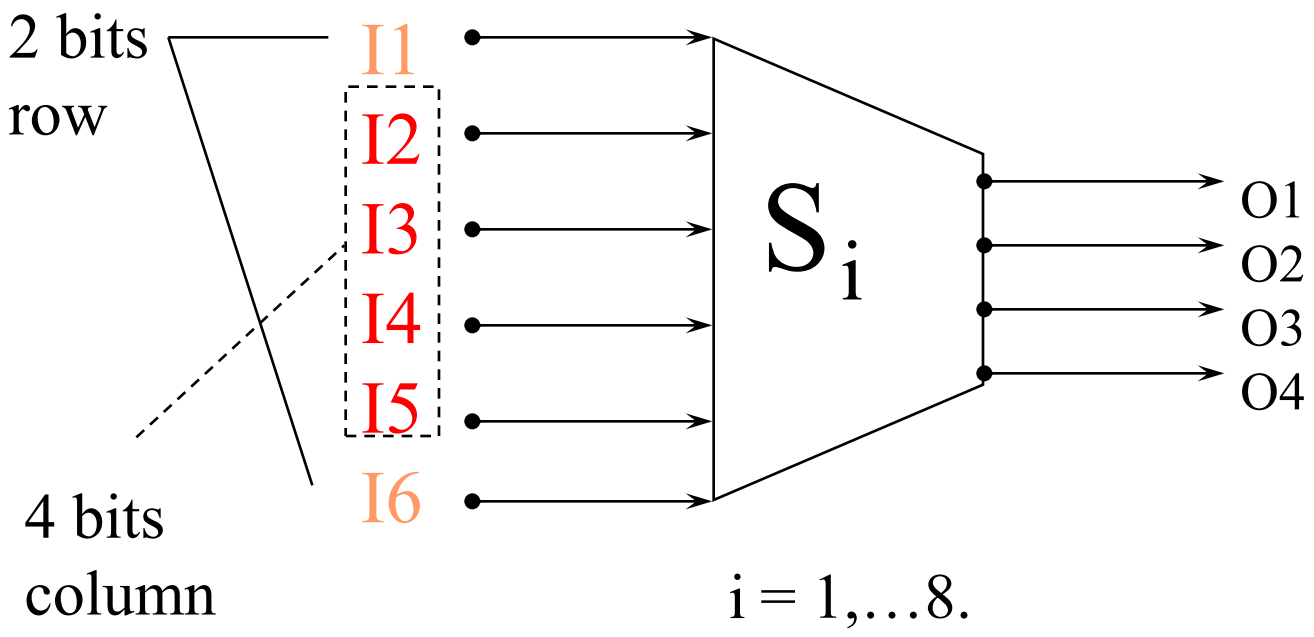
The permutation produces "spread" among the chunks/S-boxes!

Bits Expansion (1-to-m)



S-Box (Substitute and Shrink)

- 48 bits ==> 32 bits. (8*6 ==> 8*4)
- 2 bits used to select amongst 4 substitutions for the rest of the 4-bit quantity



S-Box Examples

Each row and column contain different numbers.

	0	1	2	3	4	5	6	7	8	9…. 15
0	14	4	13	1	2	15	11	8	3	
1	0	15	7	4	14	2	13	1	10	
2	4	1	14	8	13	6	2	11	15	
3	15	12	8	2	4	9	1	7	5	

Example: input: 100110 output: ???

DES Standard : Summry

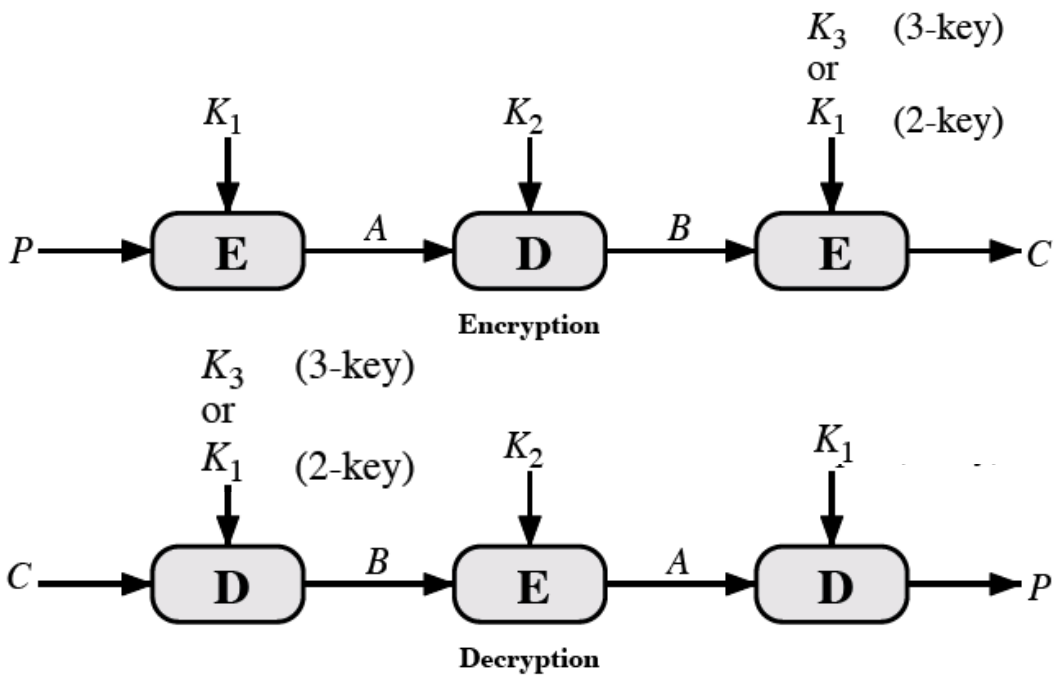
- Cipher Iterative Action :
  - Input: 64 bits
  - Key: 48 bits
  - Output: 64 bits
- Key Generation Box :
  - Input: 56 bits
  - Output: 48 bits

One round (Total 16 rounds)

3DES (Triple-DES)

- Three stages of DES
  - with two different keys
    - some attacks are possible but impractical
    - Merkle and Hellman, 1981
      - 2^{56} trials, but requires 2^{56} plaintext-ciphertext pairs
    - Oorschot and Wiener, 1990
      - $2^{120}/n$ trials, where n is the number of plaintext-ciphertext pairs
  - with three different keys
    - Attack complexity increases and becomes impractical

Triple-Des with two/three keys



(b) Triple Encryption

- E-D-E sequence
  - use of decryption at the second stage does not reduce/increase the security
  - Why decryption in the middle stage?

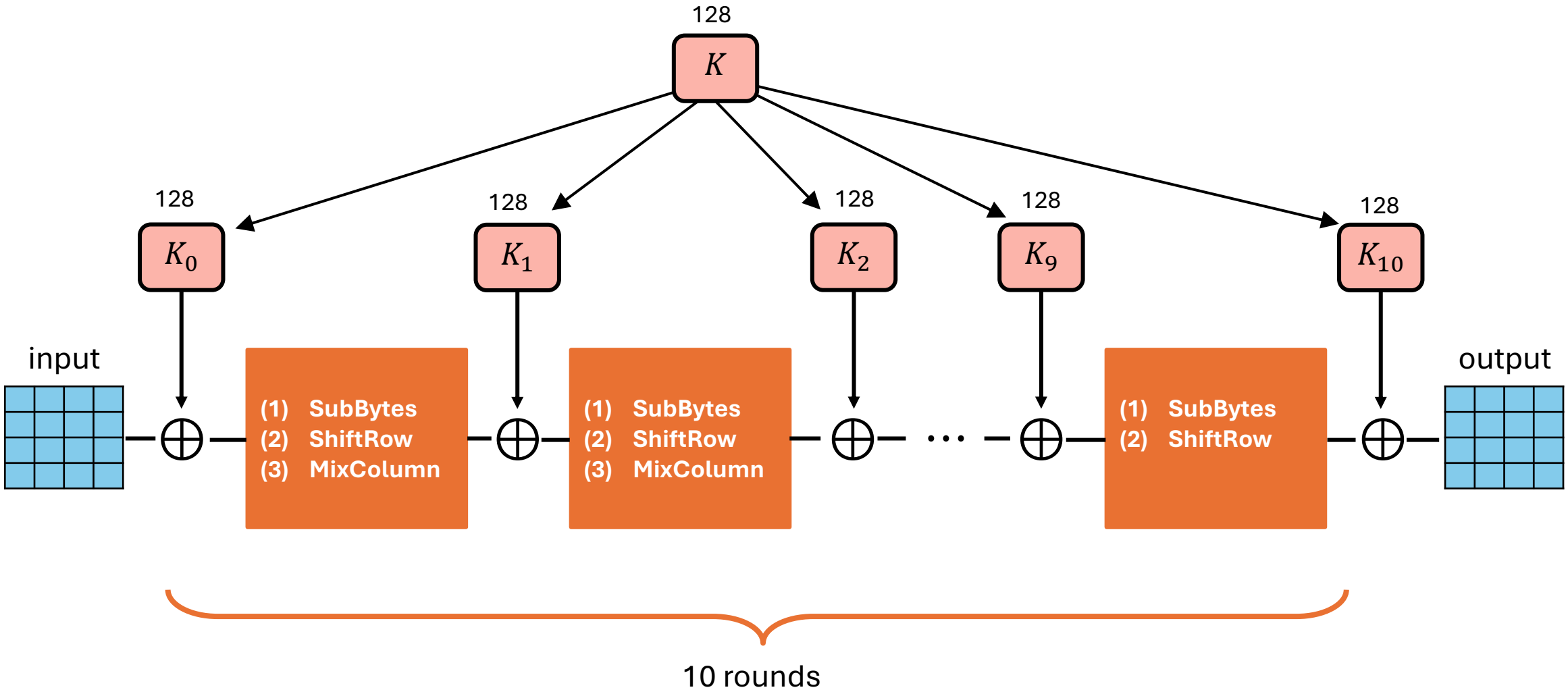
Advanced Encryption Standard(AES)

- Why DES Was Replaced
  - Key length too short (56 bits)
  - Block length too short (64 bits)
  - Too slow in software

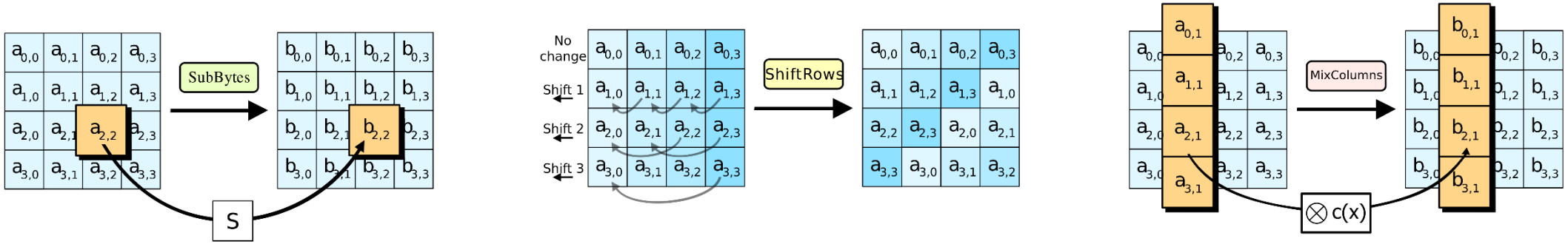
- If you remember nothing else:

  - $\text{AES-128} : \{0,1\}^{128} \times \{0,1\}^{128} \rightarrow \{0,1\}^{128}$
  - $\text{AES-192} : \{0,1\}^{192} \times \{0,1\}^{128} \rightarrow \{0,1\}^{128}$
  - $\text{AES-256} : \{0,1\}^{256} \times \{0,1\}^{128} \rightarrow \{0,1\}^{128}$

AES-128

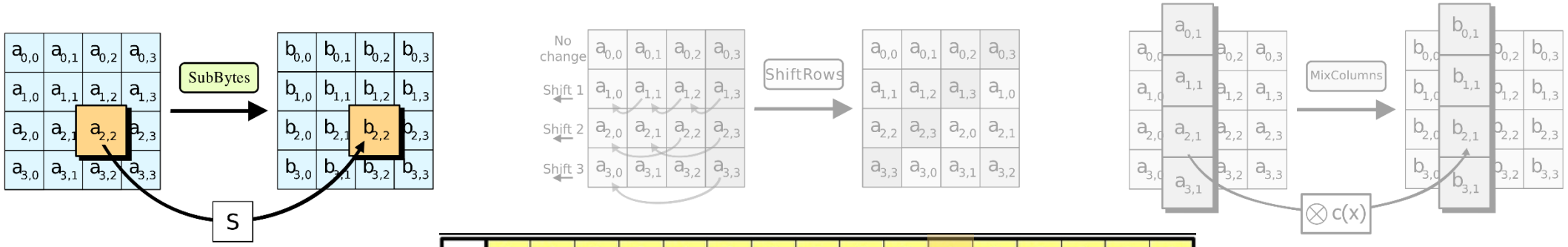

128
K
128
K_0
128
K_1
128
K_2
128
K_9
128
K_{10}
input
output
(1) SubBytes
(2) ShiftRow
(3) MixColumn
(1) SubBytes
(2) ShiftRow
(3) MixColumn
(1) SubBytes
(2) ShiftRow
10 rounds

AES round function



(1) SubBytes
(2) ShiftRow
(3) MixColumn

AES round function - SubBytes

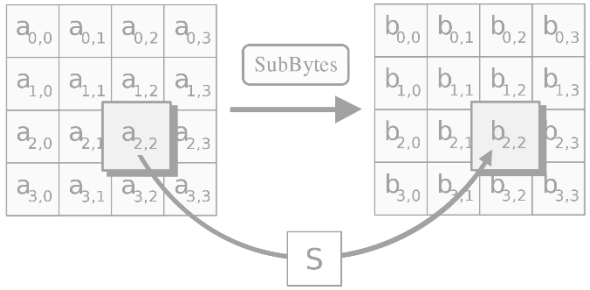


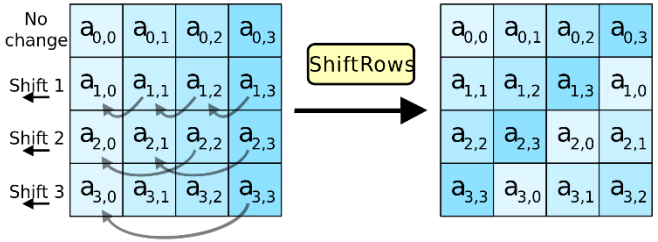
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

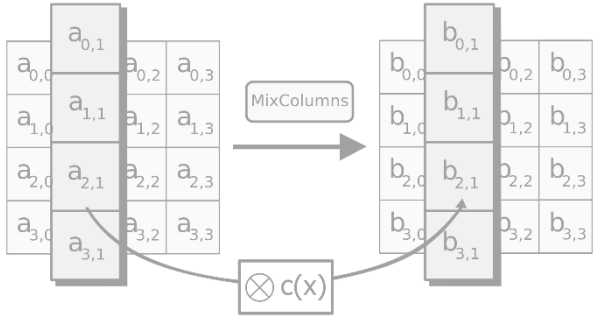
$a_{2,2} = 8\text{A}$

$b_{2,2} = 7\text{E}$

AES round function - ShiftRows

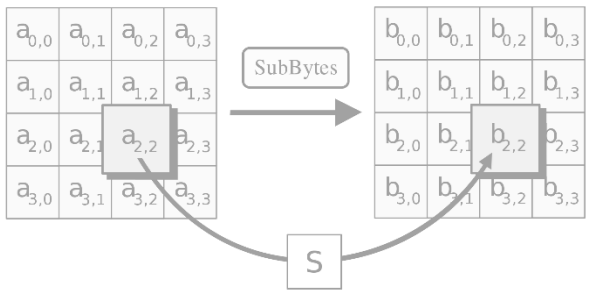


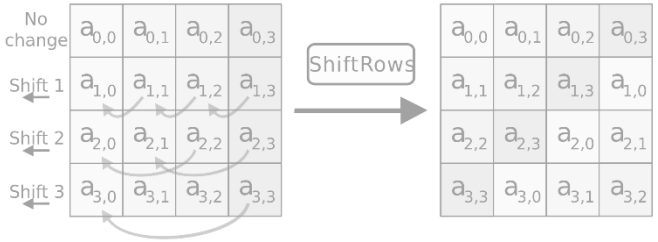


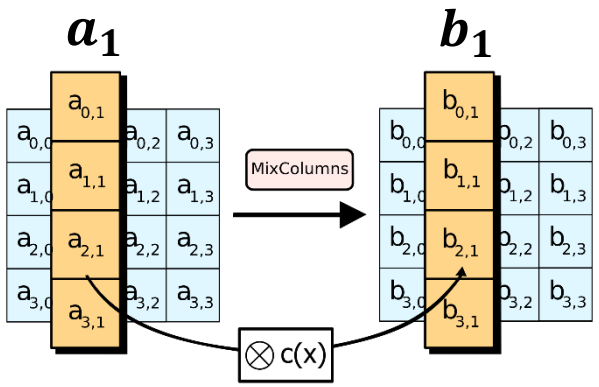


(1) **SubBytes**
(2) **ShiftRow**
(3) **MixColumn**

AES round function - MixColumns







(1) SubBytes
(2) ShiftRow
(3) MixColumn

$$M\boldsymbol{a_1} = \boldsymbol{b_1}$$

$$\begin{pmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{pmatrix} \begin{pmatrix} a_{0,1} \\ a_{1,1} \\ a_{2,1} \\ a_{3,1} \end{pmatrix} = \begin{pmatrix} b_{0,1} \\ b_{1,1} \\ b_{2,1} \\ b_{3,1} \end{pmatrix}$$

$$1 \cdot a_{0,1} + 2 \cdot a_{1,1} + 3 \cdot a_{2,1} + 1 \cdot a_{3,1} = b_{1,1}$$

$$a_{0,1} \oplus 2 * a_{1,1} \oplus 3 * a_{2,1} \oplus a_{3,1} = b_{1,1}$$

AES round

$a_{0,0}$	$a_{0,1}$	$a_{0,2}$	$a_{0,3}$
$a_{1,0}$	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$
$a_{2,0}$	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$
$a_{3,0}$	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$

SubBytes	$b_{i,j} = S[a_{i,j}]$
ShiftRows	$\begin{bmatrix} c_{0,j} \\ c_{1,j} \\ c_{2,j} \\ c_{3,j} \end{bmatrix} = \begin{bmatrix} b_{0,j} \\ b_{1,j-1} \\ b_{2,j-2} \\ b_{3,j-3} \end{bmatrix}$
MixColumns	$\begin{bmatrix} d_{0,j} \\ d_{1,j} \\ d_{2,j} \\ d_{3,j} \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} c_{0,j} \\ c_{1,j} \\ c_{2,j} \\ c_{3,j} \end{bmatrix}$
AddRoundKey	$\begin{bmatrix} e_{0,j} \\ e_{1,j} \\ e_{2,j} \\ e_{3,j} \end{bmatrix} = \begin{bmatrix} d_{0,j} \\ d_{1,j} \\ d_{2,j} \\ d_{3,j} \end{bmatrix} \oplus \begin{bmatrix} k_{0,j} \\ k_{1,j} \\ k_{2,j} \\ k_{3,j} \end{bmatrix}$

$$\begin{bmatrix} e_{0,j} \\ e_{1,j} \\ e_{2,j} \\ e_{3,j} \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} S[a_{0,j}] \\ S[a_{1,j-1}] \\ S[a_{2,j-2}] \\ S[a_{3,j-3}] \end{bmatrix} \oplus \begin{bmatrix} k_{0,j} \\ k_{1,j} \\ k_{2,j} \\ k_{3,j} \end{bmatrix}$$

$$= \left(\begin{bmatrix} 02 \\ 01 \\ 01 \\ 03 \end{bmatrix} \cdot S[a_{0,j}] \right) \oplus \left(\begin{bmatrix} 03 \\ 02 \\ 01 \\ 01 \end{bmatrix} \cdot S[a_{1,j-1}] \right)$$

$$\oplus \left(\begin{bmatrix} 01 \\ 03 \\ 02 \\ 01 \end{bmatrix} \cdot S[a_{2,j-2}] \right) \oplus \left(\begin{bmatrix} 01 \\ 01 \\ 03 \\ 02 \end{bmatrix} \cdot S[a_{3,j-3}] \right) \oplus \begin{bmatrix} k_{0,j} \\ k_{1,j} \\ k_{2,j} \\ k_{3,j} \end{bmatrix}$$

$T_0[x] = \left(\begin{bmatrix} 02 \\ 01 \\ 01 \\ 03 \end{bmatrix} \cdot S[x] \right)$	$T_1[x] = \left(\begin{bmatrix} 03 \\ 02 \\ 01 \\ 01 \end{bmatrix} \cdot S[x] \right)$	$T_2[x] = \left(\begin{bmatrix} 01 \\ 03 \\ 02 \\ 01 \end{bmatrix} \cdot S[x] \right)$	$T_3[x] = \left(\begin{bmatrix} 01 \\ 01 \\ 03 \\ 02 \end{bmatrix} \cdot S[x] \right)$

AES performance

	Throughput (my laptop)
AES-128 (in software)	0.27 GB/s
AES-128 (w/AES-NI)	3.45 GB/s

- AES is fast and secure, especially with Intel's AES-NI hardware acceleration.

Bireysel çalışma:

- Galois Counter Mode (GCM)
- Quantum cryptography

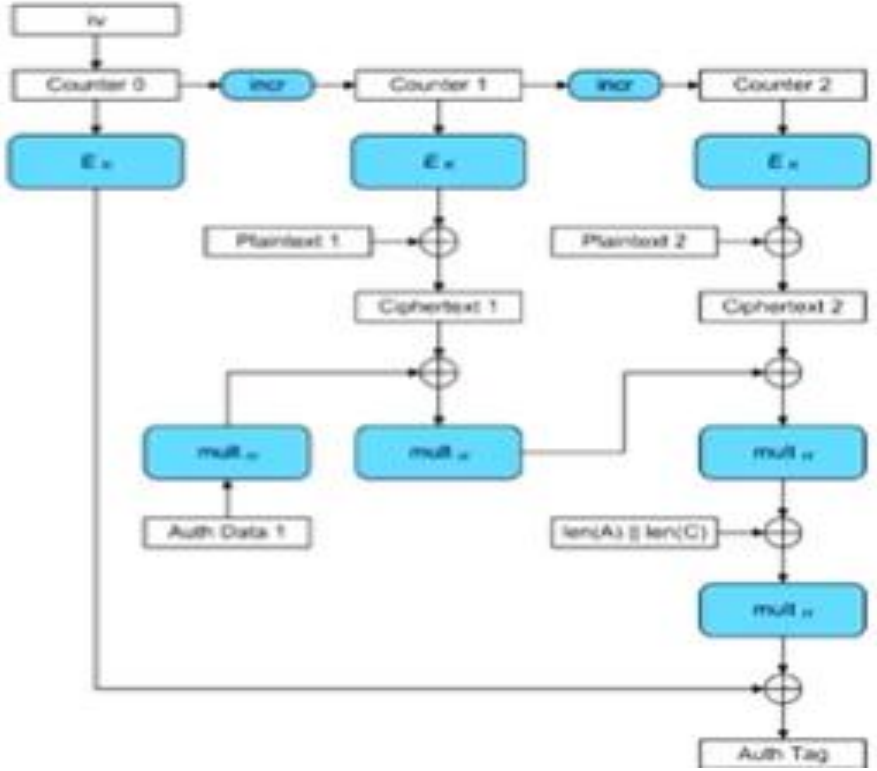
Bireysel Projeler

- OTP Uygulaması (Herkes için). Stream Cipher
- DES ve AES Uygulaması (Herkes için). Block Cipher

- **Son Teslim Tarihi (Her iki proje için):** 1 Nisan 2026.

Galois Counter Mode (GCM)

- **Derivative of the CTR mode of operation**
- **Pros:**
  - Generates a message authentication code for the message
  - Very fast and efficient
- **Cons:**
  - More difficult to understand



Quantum cryptography

How? secures data and detects eavesdroppers using quantum physics.

Advantage:

security is based on the laws of quantum physics

Disadvantage:

needs a dedicated equipment.

Practicality?

Currently: successful transmissions for distances of length around **150 km**.
Commercial products are available.

Quantum indeterminacy: quantum states **cannot** be measured without disturbing the original state.
Hence **Eve** cannot read the bits in an unnoticeable way.